



Health Claims Forum

Working together for success

The Cancer Challenge

16 July 2024

Agenda



- **Welcome** - Keiran Tarrant – Chair Health Claims Forum
- **Reclassification and Problematic Cancers** – *Professor James Brenton, Oncologist & Medical Officer for Pacific Life Re.*
- Break
- **Prostate Cancer** - *Dr Radi Counsell, Specialist in Gynaecological and Breast Cancers & Medical Officer for RGA.*
- Lunch
- **Lynn’s story** – *Lynn chats to Tim about her own cancer journey.*
- **Practical approach to support individuals with cancer back to work** - *Beth Husted, Wellbeing and Customer Experience Manager, Unum.*
- Break
- **Paddling in invested waters? Cancer screening – should we do it?** - *Dr John O’Brien, Chief Medical Officer for Gen Re.*
- **Networking - Slug and Lettuce** – over the road.



Health Claims Forum
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Reclassification and Problematic Cancers

Professor James Brenton



PACIFIC LIFE RE

Cancer classification

Professor James D. Brenton PhD FRCP
University of Cambridge and Cambridge
University Hospitals NHS Foundation Trust

16 July 2024

Health Claims Forum – The Cancer Challenge

16 July 2024

Objectives

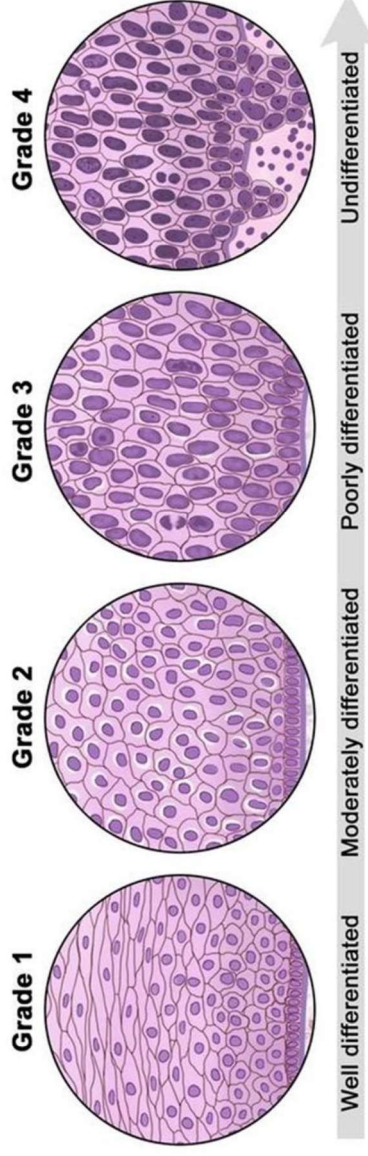
- Understand common terms for cancer critical illness and their limitations
- Discuss the process for cancer classification by the World Health Organization (WHO)
- Review recent changes to the classification of pituitary neoplasms
- Discuss challenges regarding critical illness claims for neuroendocrine neoplasms

How do we define cancer?



Cancer is largely defined by cellular appearance

Tumor Grade

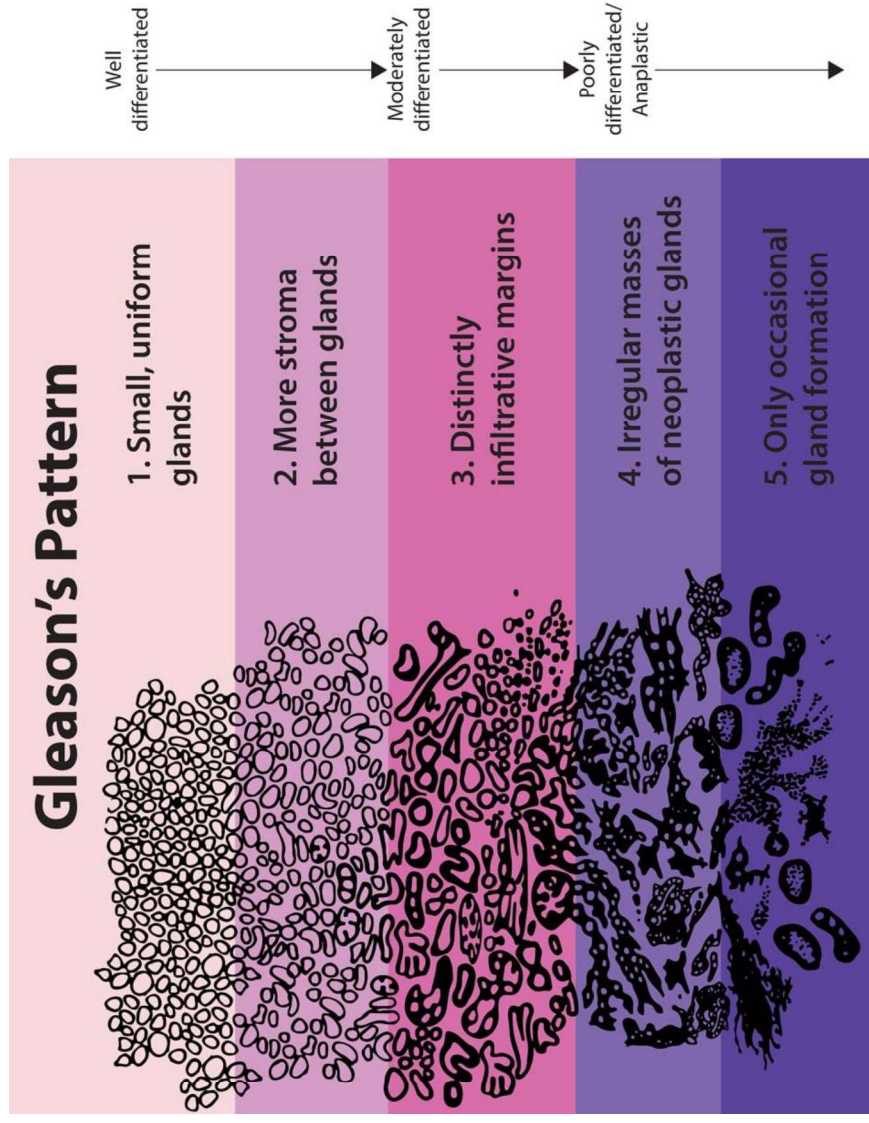


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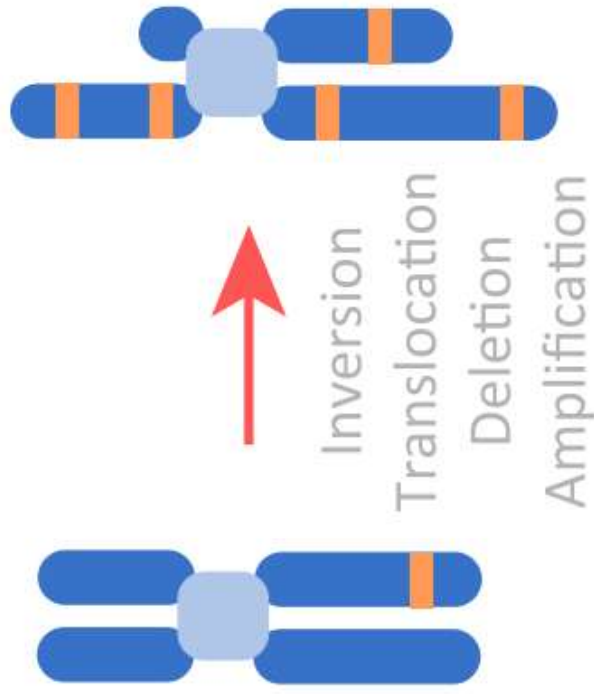
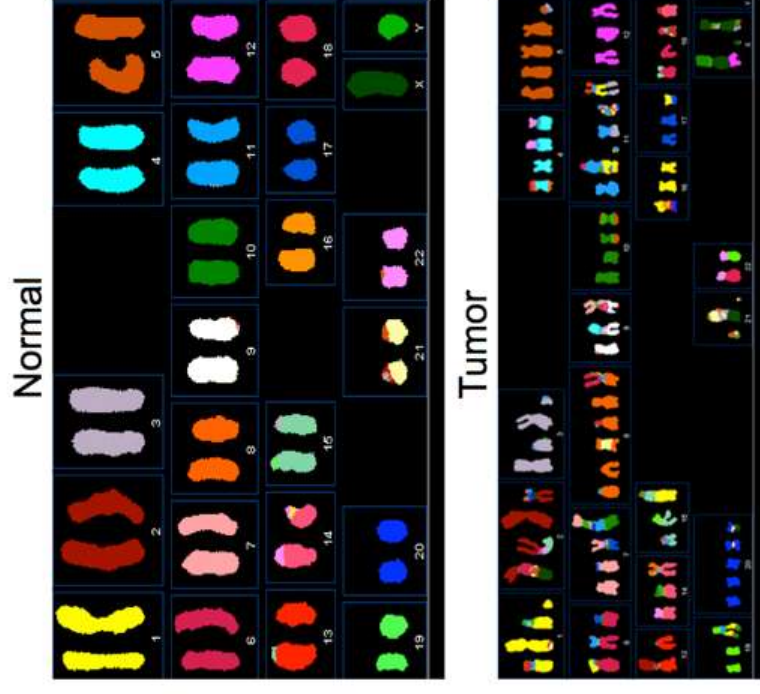
Grade

Grade is determined by examination by a pathologist and is a measure of how similar cancer cells are to their cell of origin. This is often called **differentiation**. Key features include the ratio of nuclear size to cellular size and relationship to surrounding normal cells.

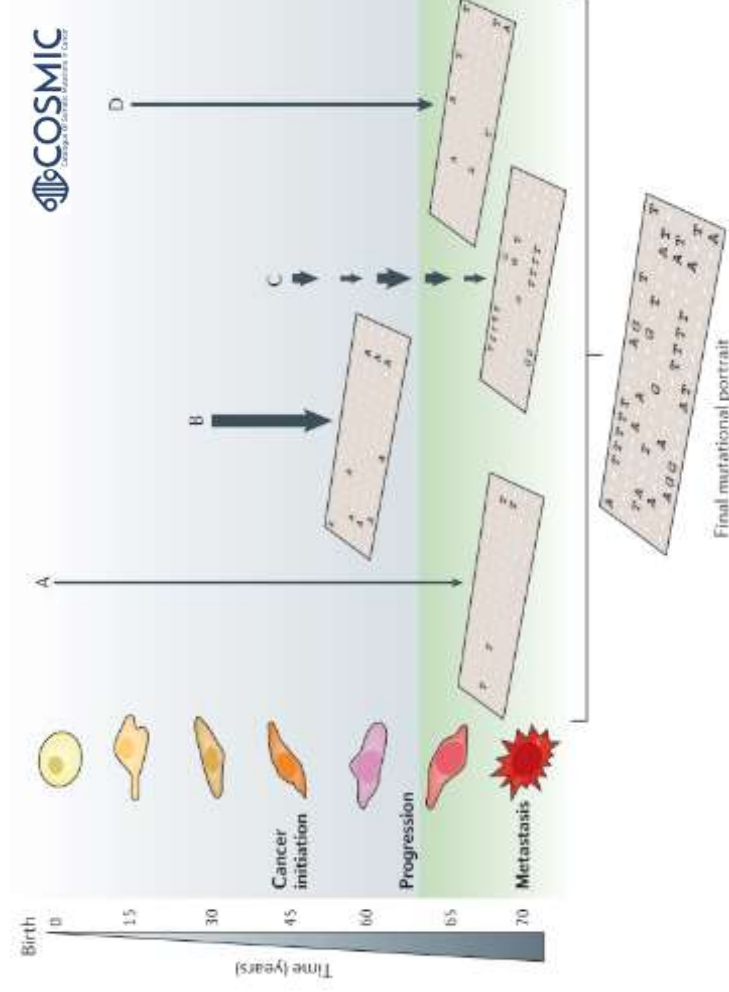
Gleason grade for prostate carcinoma



Cancers arise from genetic damage to DNA

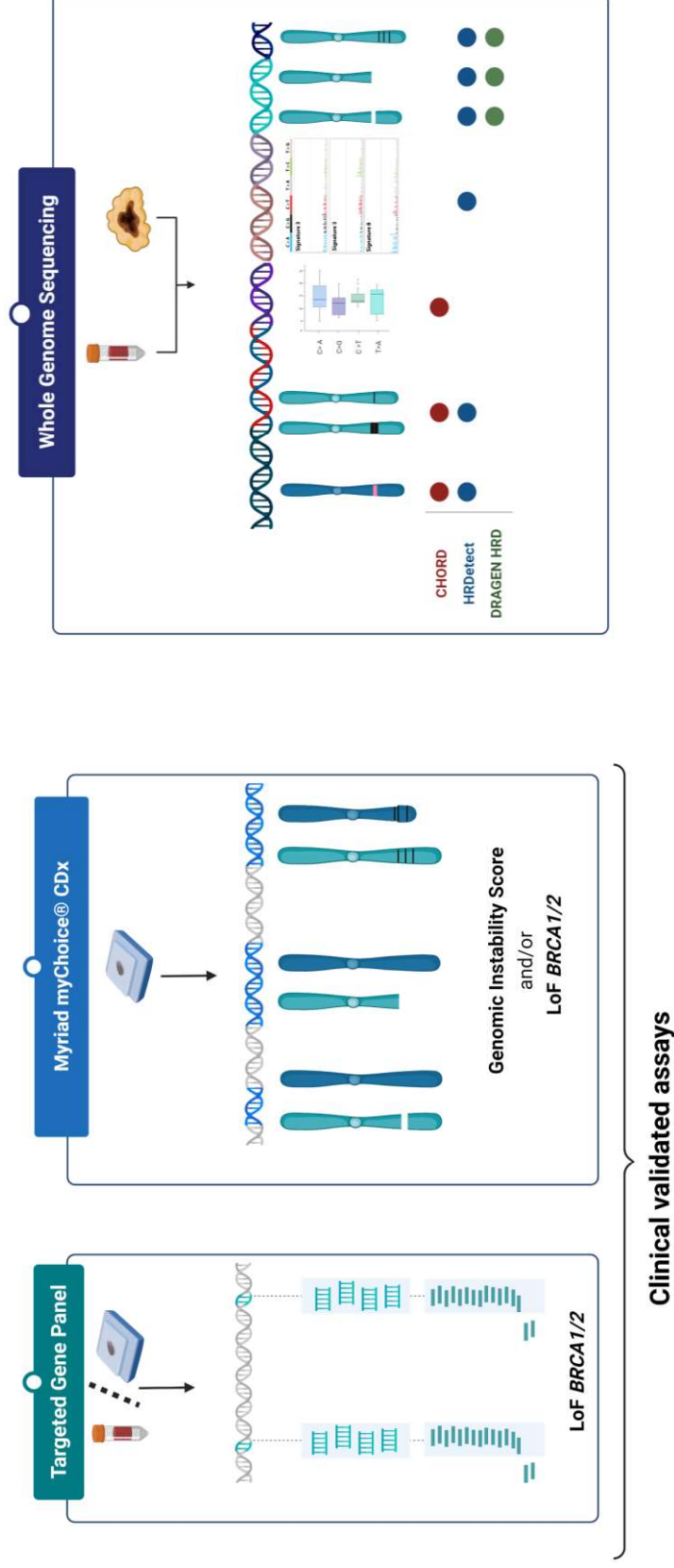


Cancer risk reflects ongoing DNA damage over life

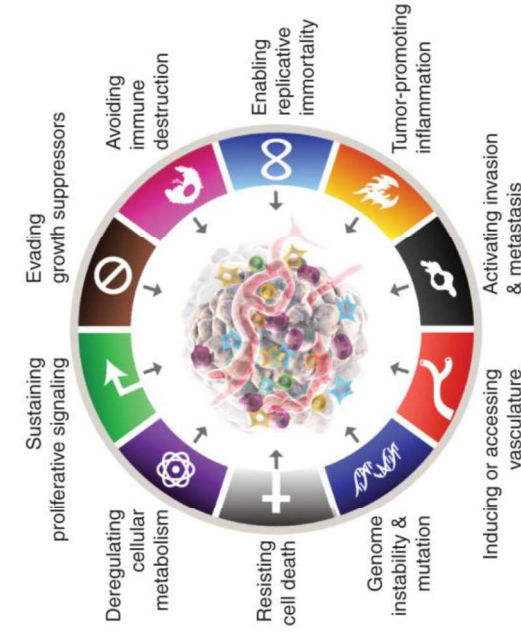


- Mutations are an *archaeological* record of exposure to mutational processes
- Signatures are the *patterns* of mutational processes
- Individual cancers have *multiple* mutational signatures

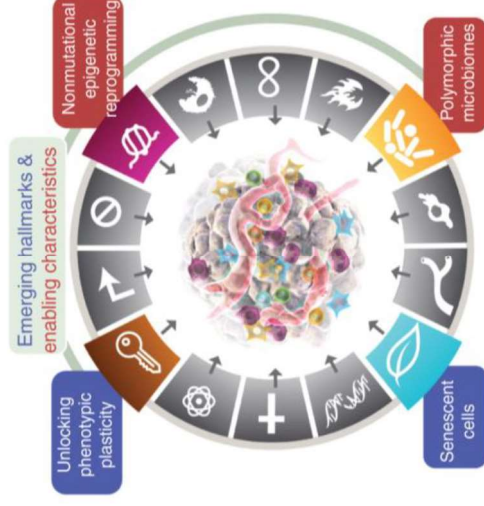
Clinical testing includes cancer sequencing



Cancer behaviour is complex



2000



2020

Hallmarks of cancer TNM classification

The hallmarks are fundamental properties of cancer that provide principles to define complex cancer **phenotypes**

Ongoing research identifies new cancer mechanisms which are periodically included

ABI guide to minimum standards for CI cover (2022)



Cancer definitions

Cancer

Any malignant tumour positively diagnosed with histological confirmation and characterised by the uncontrolled growth of malignant cells and invasion of tissue.

The term malignant tumour includes leukaemia, sarcoma, and lymphoma except those that arise from or are confined to the skin (including cutaneous lymphomas and sarcomas).

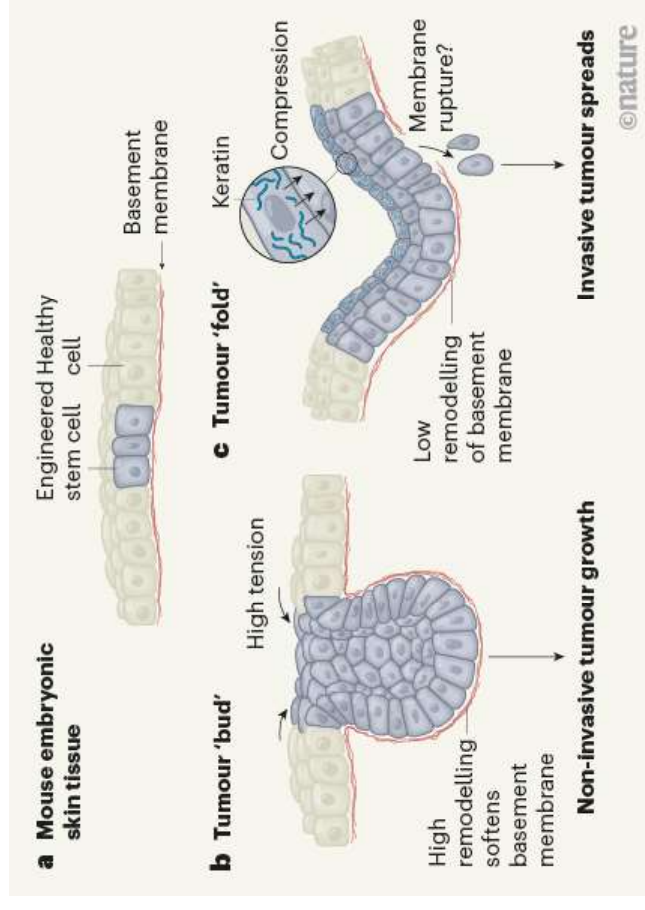
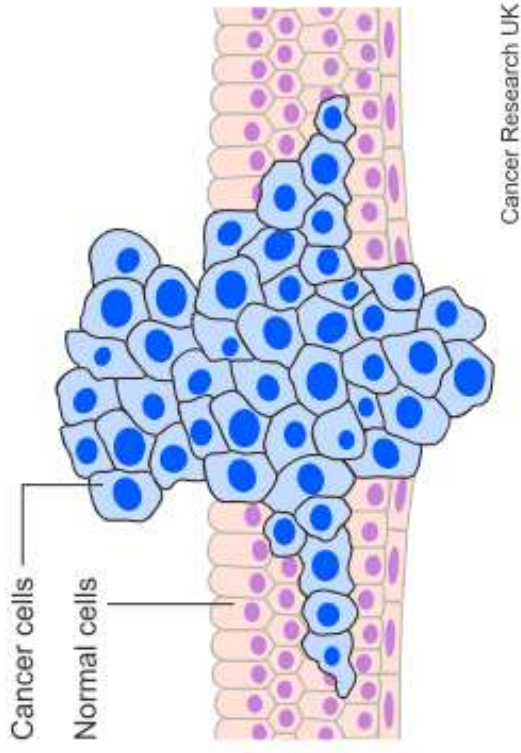
Borderline malignancy

Potentially malignant cells that have not invaded the adjacent tissue.

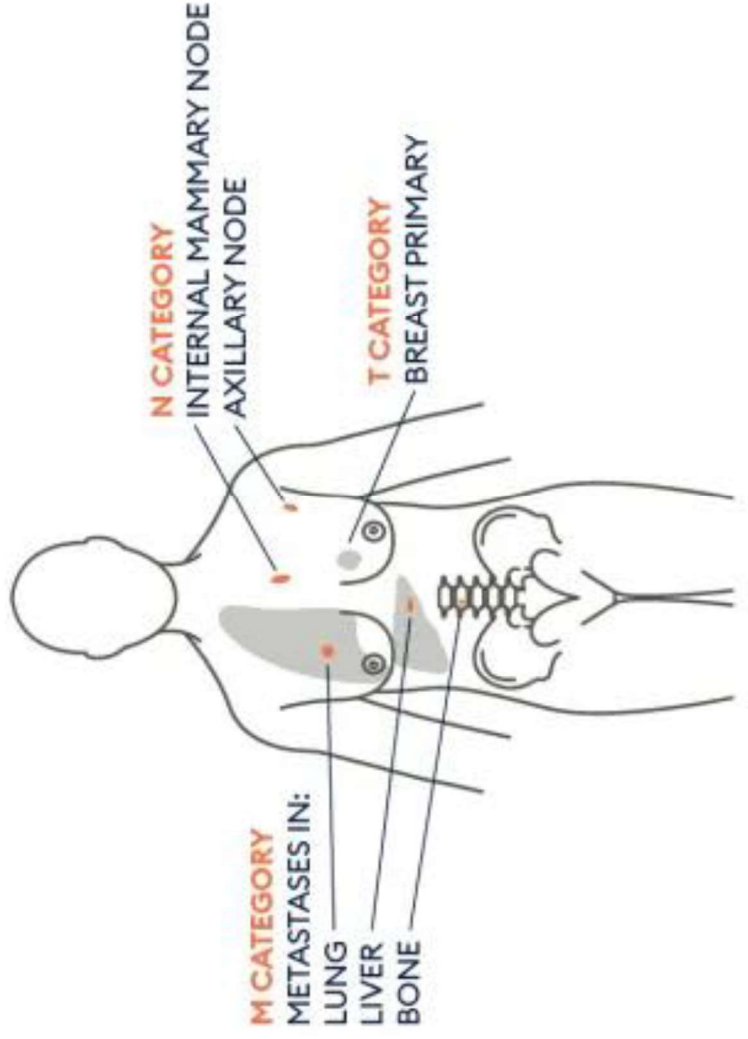
Invasion

In relation to cancer means the occurrence of malignant/cancerous cells that have spread beyond the layer of tissue in which it developed and is growing into surrounding, healthy tissues made up of different cells (that is, more extensive than cancer in-situ).

Invasion is pathologically defined



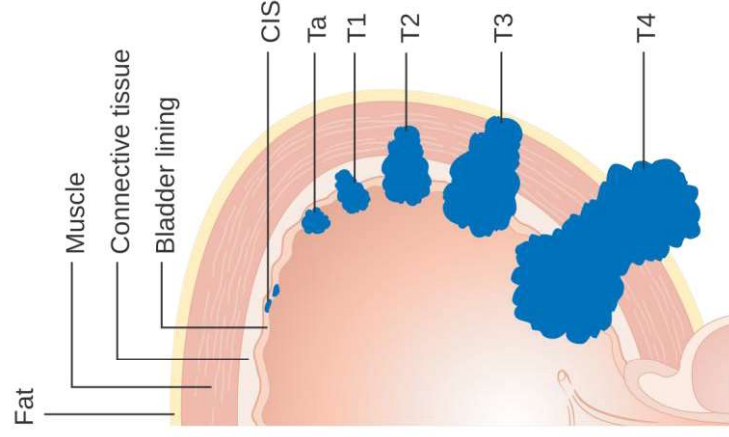
Cancer staging is clinically based



UICC TNM classification

The TNM classification is an anatomically based system that records the primary and regional nodal extent of the tumour and the absence or presence of metastases.

UICC TNM classification for the urinary bladder



T categories

Physical examination,
imaging and endoscopy

N and M categories

Physical examination and
imaging



Non-invasive follicular thyroid neoplasm with papillary-like nuclear features

Definition

Non-invasive follicular thyroid neoplasm with papillary-like nuclear features (NIFTP) is a non-invasive encapsulated/well demarcated follicular cell derived tumour with a follicular growth pattern and nuclei resembling papillary thyroid carcinoma (PTC) that has an extremely low malignant potential.

ICD-O coding

8349/1 Non-invasive follicular thyroid neoplasm with papillary-like nuclear features

ICD-11 coding

2F9A & XA8RK3 Neoplasms of unknown behaviour of endocrine glands & thyroid gland tumour, NOS

Related terminology

Not recommended: non invasive follicular variant papillary thyroid carcinoma

Subtype(s)

Subcentimeter NIFTP: <10 mm NIFTP

Oncocytic NIFTP: NIFTP with at least 75% oncocytic cells

Localization

Either thyroid lobe or the isthmus.

Clinical features

The patients usually present with an asymptomatic nodule detected during routine physical examination or by imaging. By ultrasonography, NIFTP typically appears as a solid mildly hypoechoic or isoechoic mass, round to ovoid in shape with smooth borders and lacking calcifications ([29317047](#)).

△ ▲ A ■ ■ ■

Definition

ICD-O coding

ICD-11 coding

Related terminology

Subtype(s)

Localization

Clinical features

Epidemiology

Etiology

Pathogenesis

Macroscopic appearance

Histopathology

Cytology

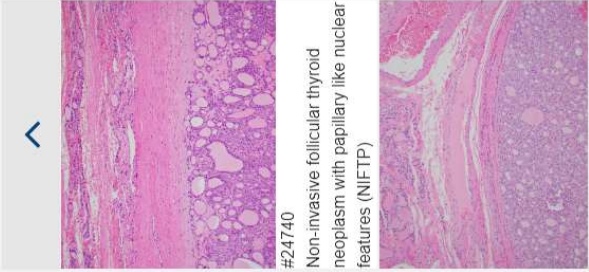
Diagnostic molecular pathology

Essential and desirable diagnostic criteria

Staging

Prognosis and prediction

Add Personal Note



#24740

Non-invasive follicular thyroid neoplasm with papillary like nuclear features (NIFTP)

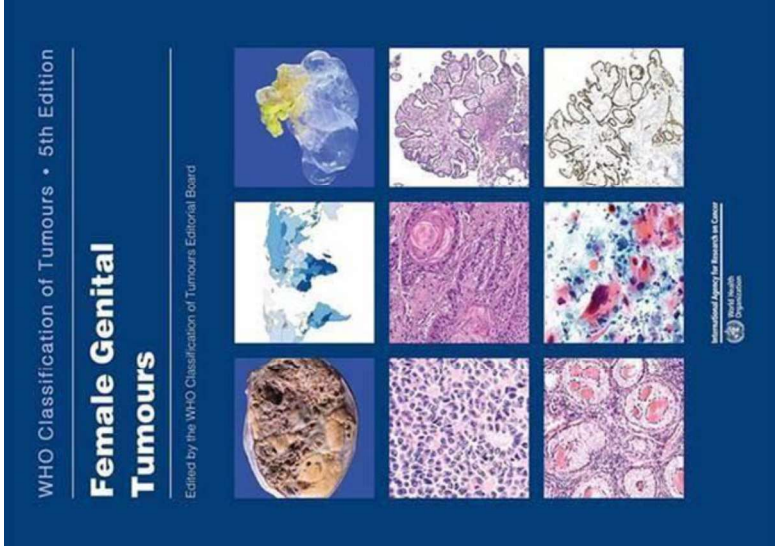
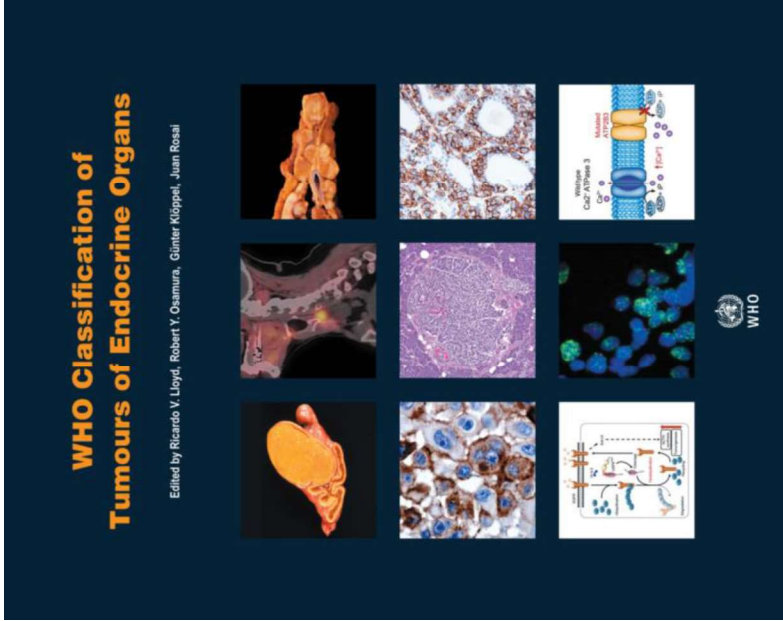
Other pitfalls

- Borderline ovarian neoplasms often are staged with FIGO/TNM staging but are non-invasive
- Low-grade appendiceal mucinous neoplasm (LAMN); also termed pseudomyxoma peritonei when advanced should be T4 if acellular mucin is present on serosa
- Endometrial carcinoma is T1a if limited to endometrium or <50% of myometrium

How do we classify cancer?



WHO “Blue books”



About Us

The definitive diagnosis and classification of individual cancers underpins the care of individual cancer patients, as well as research into cancer causation, prevention, diagnosis, and treatment. Traditionally, cancer classification has been based on consensus of histopathological opinion, with very limited consideration of molecular pathology. But new technologies are now transforming the field of pathology more rapidly than at any other time during the past 30 years, and it has become increasingly clear that the traditional approach to cancer classification is insufficient. Our understanding of cancer at the molecular level has now reached the point that this information must be included in diagnoses. Digital pathology and image analysis are also producing new insights, providing quantitative information that many existing diagnostic criteria while challenging others. The rapid improvement in computer technology, artificial intelligence, is already producing clinically applicable aids to diagnosis, and this trend is likely to continue.

IARC has been responsible for the WHO Classification of Tumours, also known as the WHO Blue Books, since its first edition (2000–2005), which covered all organ sites in 10 volumes. The WHO Classification of Tumours is the regular basis. The characteristics of each cancer type, including diagnostic criteria, pathological features, and molecular alterations, are described and illustrated in a strictly disease-oriented manner to provide standards for diagnosis and cancer research. The new website represents a new departure for the WHO Classification of Tumours and facilitates many innovations, including the use of whole slide images.

None of this would be possible without the contributions of some 200 authors and editors who made the WHO Classification of Tumours what it is today. We are very grateful to all of them for their contributions.



Example classification



International Agency for Research on Cancer
World Health Organization

WHO Classification of Tumours online

Home Account Notes Favourites Search About Contact Logout

Central Nervous System Tumours (5th ed.) > Tumours of the sellar region > Pituitary adenoma / pituitary neuroendocrine tumour >

Definition

Pituitary adenoma / pituitary neuroendocrine tumour (PitNET) is a clonal neoplastic proliferation of anterior pituitary hormone-producing cells.

ICD-O coding

8272/3 Pituitary adenoma / pituitary neuroendocrine tumour (PitNET)

ICD-11 coding

2F37.Y & XH94U0 Other specified benign neoplasm of endocrine glands & Pituitary adenoma, NOS
2F9A & XH94U0 Neoplasms of unknown behaviour of endocrine glands & Pituitary adenoma, NOS

Related terminology

Acceptable: PitNET; pituitary adenoma.

Subtype(s)

The types and subtypes of pituitary adenomas / PitNETs are described in #29187Table 12.01.

Localization

These tumours are usually identified in the sellar region, often with suprasellar extension, but ectopic locations include the sphenoid sinus, and rare clival and suprasellar tumours have also been described [30895500]. Rarely, pituitary adenomas / PitNETs may arise in teratomas [24564638; 3548446; 3263841].

Clinical features

Pituitary adenomas / PitNETs have a spectrum of clinical features. They may be small slow-growing and found incidentally, or they may cause rise in hormone excess syndromes, infiltration by...

View tables and boxes

#16978

Transcriptome of pituitary neuroendocrine tumours (PitNETs)

16 July 2024

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Paediatric brain tumours are molecularly defined



nature medicine



Article

<https://doi.org/10.1038/s41591-023-02255-1>

Multimic neuropathology improves diagnostic accuracy in pediatric neuro-oncology

Received: 4 August 2022

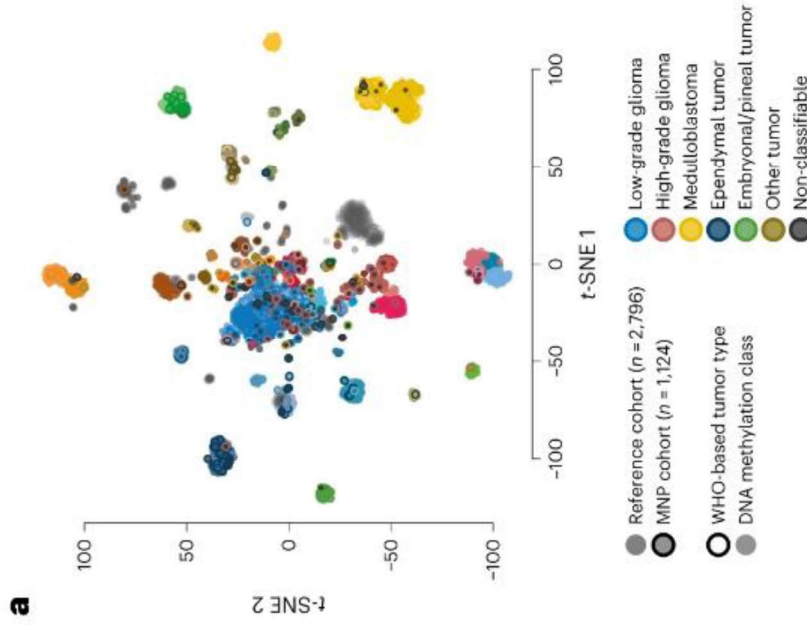
Accepted: 13 February 2023

Published online: 16 March 2023

Check for updates

A list of authors and their affiliations appears at the end of the paper

The large diversity of central nervous system (CNS) tumor types in children and adolescents results in disparate patient outcomes and renders accurate diagnosis challenging. In this study, we prospectively integrated DNA methylation profiling and targeted gene panel sequencing with blinded neuropathological reference diagnostics for a population-based cohort of more than 1,200 newly diagnosed pediatric patients with CNS tumors, to assess their utility in routine neuropathology. We show that the multi-omic integration increased diagnostic accuracy in a substantial proportion of patients through annotation to a refining DNA methylation class (50%), detection of diagnostic or therapeutically relevant genetic alterations (47%) or identification of cancer predisposition syndromes (10%). Discrepant results by neuropathological WHO-based and DNA methylation-based classification (30%) were enriched in histological high-grade gliomas, implicating relevance for current clinical patient management in 5% of all patients. Follow-up (median 2.5 years) suggests improved survival for patients with histological high-grade gliomas displaying lower-grade molecular profiles. These results provide preliminary evidence of the utility of integrating multi-omics in neuropathology for pediatric neuro-oncology.



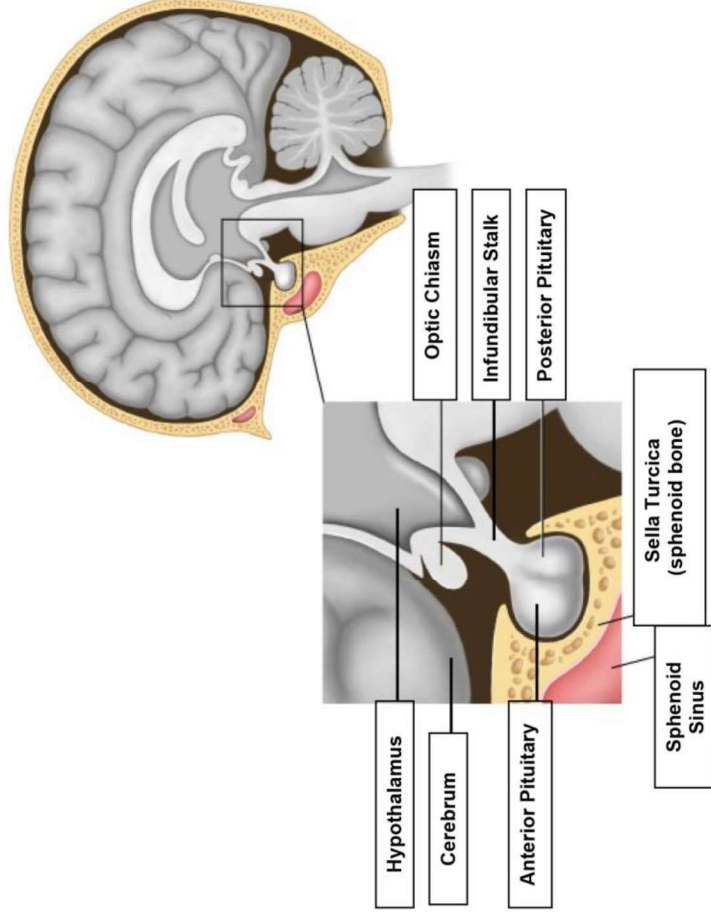
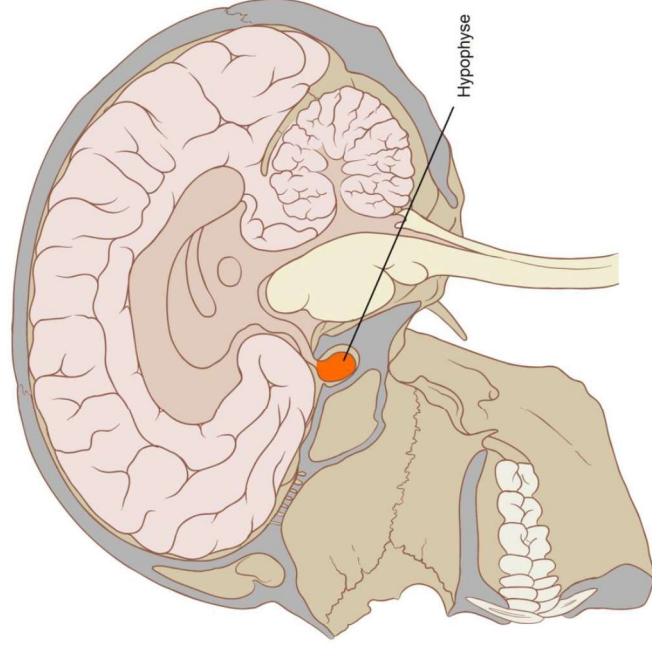
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Why were neuroendocrine neoplasms reclassified?



Anatomy of the pituitary gland



Classification of neuroendocrine neoplasms (NEN)



Motivation

- **Previous descriptions were highly variable.** Different site-specific terminologies and criteria. Confusing for pathologists and cancer clinicians
- **Need for uniform classification framework.** For NEN at **any** anatomical location
- **Consensus meeting November 2017 with recent publications**

Pituitary neoplasms are now neuroendocrine tumours:

- The 2022 WHO classification of Endocrine Tumours and Central Nervous System tumours **reclassified pituitary adenomas as neuroendocrine tumours (NETs)**
- The difficulty of specifically predicting the behavior of well-differentiated NENs is well-known, and although organ-specific grading schemes have aided in stratifying relative aggressiveness, the **proposed conceptual terminology expressly avoids categorizing the neoplasms as explicitly “benign” or “malignant”**.
- There is **no intent** for the designation of NET to affect the current understanding of malignant potential, which should remain an organ-specific characteristic.



WHO classification of tumours of the pituitary

Pituitary adenomas	Somatotroph adenoma
	Lactotroph adenoma
	Thyrotroph adenoma
	Corticotroph adenoma
	Gonadotroph adenoma
	Null-cell adenoma
	Plurihormonal and double adenomas
Pituitary carcinoma	
Pituitary blastoma	
Tumors of the posterior pituitary	Pituicytoma
	Granular cell tumor of the sella
	Spindle cell oncocyoma
	Sellar ependymoma
	Gangliocytoma and mixed gangliocytoma-adenoma
Neuronal and paraneuronal tumors	Neurocytoma
	Paraganglioma
	Neuroblastoma
Craniopharyngioma	Adamantinomatous craniopharyngioma
	Papillary craniopharyngioma

What are the challenges of incidental diagnosis?



- "Pituitary incidentaloma" refers to a previously unsuspected lesion that is detected on an imaging study performed for reasons other than pituitary symptoms or disease
- The frequency of incidentally discovered signal abnormalities (<10 mm) varies from 4-20 percent by CT and 10-38 percent by MRI
- Not all sellar masses are pituitary NET but frequency is about 10% but almost all are <10mm and 63% have no change over sequential imaging
- Recommended management for incidentaloma <10mm
 - Measure only serum prolactin if there is no clinical suspicion of hormonal hypersecretion.
 - For patients with lesions 2 to 4 mm in diameter, do not perform further imaging tests.
 - For lesions between 5 to 9 mm in diameter, perform MRI yearly for two years; if the lesion is stable, the frequency can be decreased to every few years and then less often

Pituitary incidentalomas > 10mm

- Evaluation for hormonal hyper- and hyposecretion (hypopituitarism).
- For patients with gonadotroph adenomas and other clinically nonfunctioning lesions causing visual impairment or other neurologic symptoms → transsphenoidal surgery for initial therapy.
- For patients with pituitary lesions ≥ 10 mm and symptoms related to clinically important hormonal hypersecretion (hyperprolactinemia, Cushing syndrome, acromegaly) → treat appropriately with hormonal therapy, transsphenoidal surgery and/or pituitary irradiation
- For patients with pituitary lesions ≥ 10 mm without neurologic symptoms or hormonal hypersecretion → monitor with hormonal testing (for both hyper- and hyposecretion), visual testing, and pituitary MRI at 6 and 12 months during the first year
- For patients with pituitary lesions ≥ 20 mm and visual impairment or other neurologic symptoms, treatment to reduce the size of the neoplasm is **urgently indicated**. However, if there are no neurologic symptoms and the patient is an older adult and/or a poor surgical risk, then observation is appropriate

Critical illness for pituitary NET

- Will likely require pathological confirmation to fulfil definition
- Low risk lesions are unlikely to be supportable



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Prostate Cancer

Dr Radi Counsell, CMO RGA



Prostate Cancer Pitfalls in Staging Impact for CI claims

Dr Radi Counsell CMO
July 2024

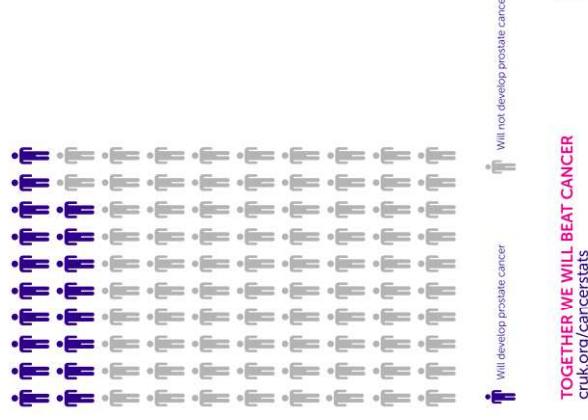


Prostate Cancer in UK

- **Most common cancer in men**
- About 55,100 new cases each year
- Incidence rate up by 48% since 1990s
- Projected rise by 15% in next 15 yr
- Increase with age: one third >75 yr age
- Autopsy occult cancer 80% age 70–90
- High risk groups:
 - African descent (lower Asians)
 - First degree relative <age 65
 - Genetic syndromes: BRCA+ Lynch

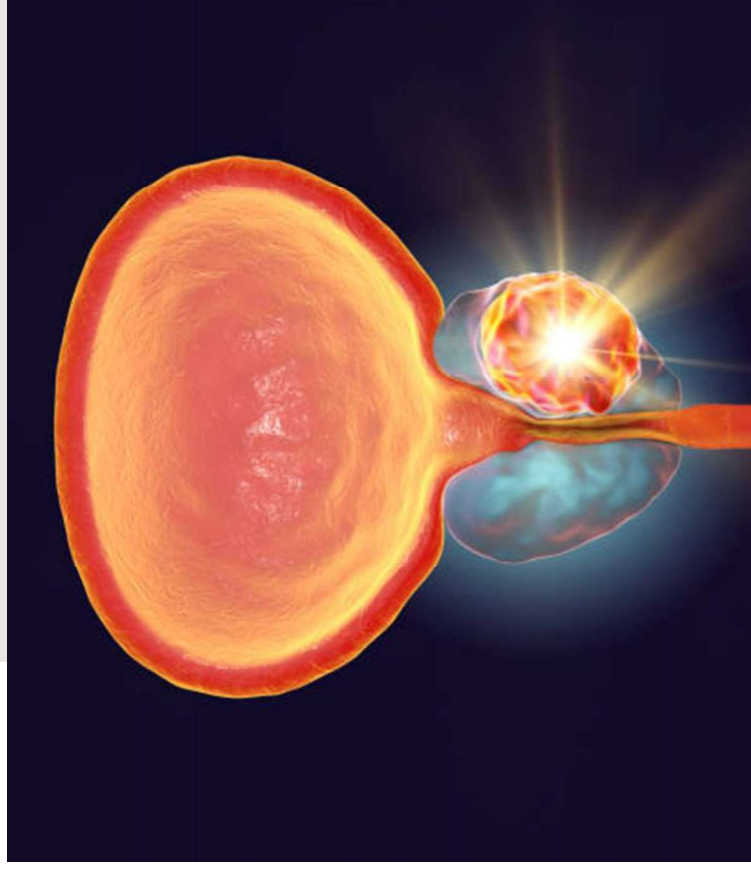
- **Risk increases with age**

Estimated Risk of Developing Prostate Cancer (ICD-10 C61) in Lifetime, Men Born After 1960, UK



Agenda

- 1 Diagnosis
- 2 MRI and PI-RADS
- 3 Staging: clinical vs pathological
- 4 Risk categories
- 5 Treatment options



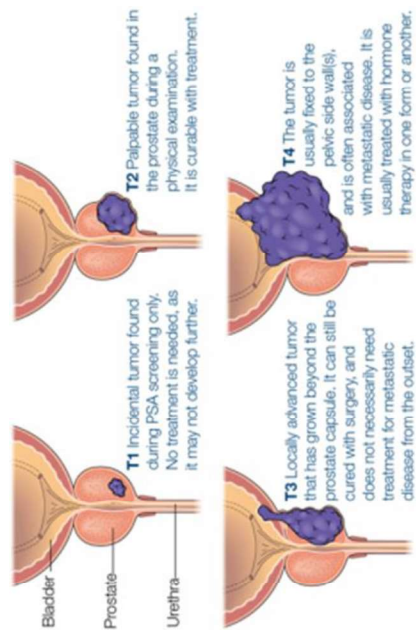
ABI definition CI claims Prostate Cancer

Time Line

GS>6 or T2 N0 M0

Clinical T stage

Date	Minimum
2000	Invasive cancer
2003	T2
2007	cT2
2012	cT2
2015	T2b
2024	cT2b or pT2



Diagnosis



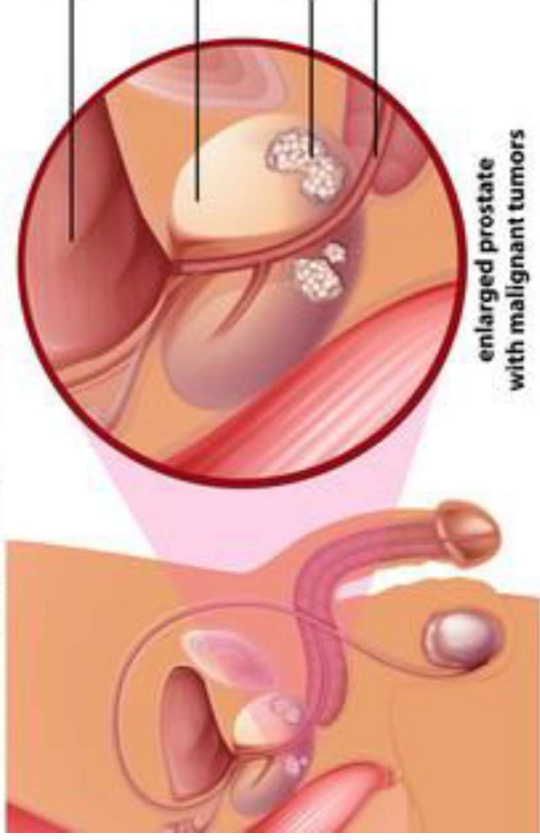
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Diagnosis Prostate Cancer

Clinical Presentation

Prostate Cancer

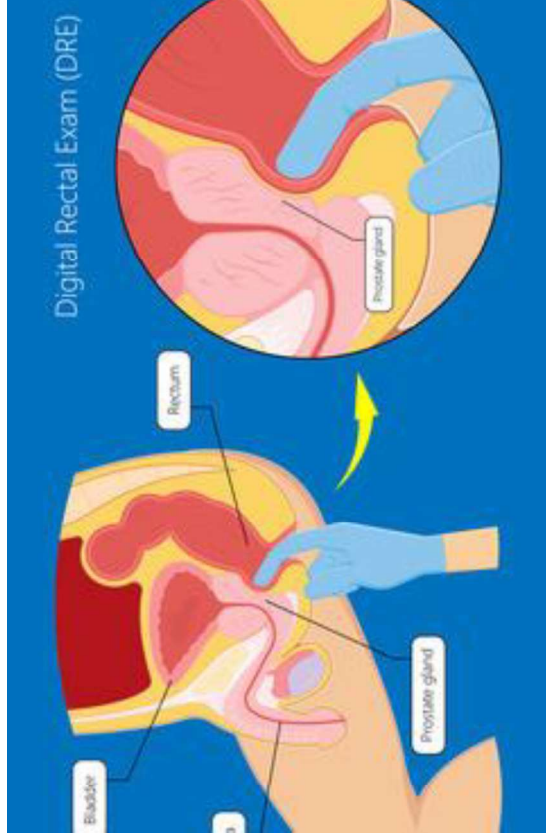


- Majority start in outer part of gland
- Early stages no symptoms
- Urinary symptoms when spread towards urethra or bladder base
- PSA by GP or Health Check
- PSA unreliable for screening ?MRI
- Majority diagnosed with cancer localised to prostate +/- nodes
- Metastatic cancer at diagnosis in 6%

Investigations

AJCC 8th Edition and UICC/TNM

Clinical assessment by DRE only



Pros/cons for cancer detection

- Easy to perform DRE
- Cancer develops outer part of gland
- Cannot assess anterior part of gland
- Seminal vesicles anterior + bladder base
- US guided biopsy under LA via transrectal or transperineal approach
- MRI better for targeted biopsies (NICE)

Prostate cancer grading and prognosis

NICE

International Soc Urological Path

Gleason Score	ISUP Grade Group	Cancer Cell vs Normal
3+3 = 6	1	Very similar
3+4 = 7	2	Most similar
4+3 = 7	3	Less like
4+4 = 8	4	Some abnormal
4+5 = 9 5+4 = 9 5+5 = 10	5	Very abnormal

Cambridge Prognostic Group

CPG	ISUP	PSA	T
1	1 And	<10 And	T1 or 2
2	2 Or	10-20 And	T1 or 2
3	2 And 3 And	10-20 And Any PSA	T1 or 2 T1 or 2
4	4 Or	>20 Or	T3
5	4 5 Any	>20 Any Any	T3 Any T4

MRI



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BENEFITS OF MRI



- Variable availability + quality reporting
- Greater use in Western world
- Endorsed in European Guidelines
 - National Institute Clinical Excellence
 - European Association of Urology
- Encouraged in US Guidelines – NCCN
- Guiding biopsy
- Cancer stage – beyond gland + nodes
- Planning treatment

Standardised Reporting for MRI

- PI-RADS Prostate Imaging Reporting and Data System used in Europe and US
- European Society of Urogenital Radiology 2012 revised 2019
- UK Centres prefer LIKERT – more subjective as considers prostate background
- 5 levels
 - 1 Highly unlikely
 - 2 Unlikely
 - 3 Equivocal
 - 4 Likely
 - 5 Very likely
- Validated with biopsy results 69% PI-RADS 4 and 94% PI-RADS 5
- LIKERT predicts more clinically significant cancers and lower false positive when used by experts

Valid CI claim?

- 54 yr old man
- PSA rise from 2.9 to 5.4 – higher than normal range (0–4 nanogram/ml)
- DRE (digital rectal examination) benign gland
- MRI 1cm lesion in left mid gland PI–RADS 3 or uncertain significance
- Biopsy positive both lobes: RIGHT ant mid and post; LEFT ant
- Surgeon gave stage as pT2c as pathology confirmed cancer in both lobes
- Treatment iodine seeds whole prostate: brachytherapy

Staging

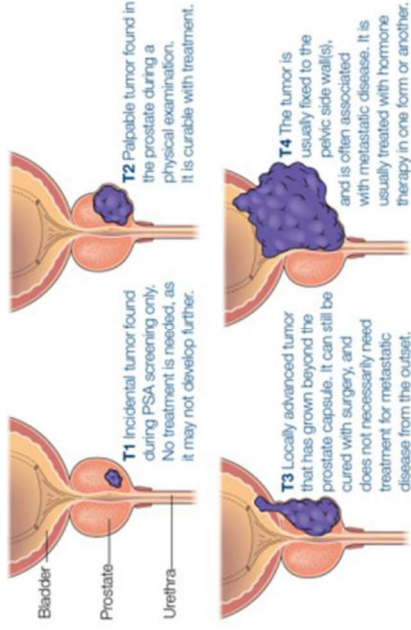


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TNM staging

Clinical Stage cT1-4



Pathological stage pT2-4 (No pT1)

- Following prostatectomy
- pT2 organ confined
- pT3 Beyond prostate
 - pT3a outside prostate or bladder neck
 - pT3b seminal vesicles
- pT4 Fixed or invasion adjacent tissues
- More accurate than clinical staging.....

Clinical vs Pathological Stage

More advanced pT stage linked to PSA

SEER study 2010 – 2011

- Nearly 29,000 men
- cT1/2 GS 6/7
- Radical prostatectomy
- pT stage was higher than cT stage
- Related to GS and PSA i.e. higher GS or PSA was linked to upstaging

Results

	GS 6	GS 7
PSA <10	16%	28%
PSA 20-20.99	39%	49%

Tumour upstaging with surgery

Case Study clinical T1 but pathological T2

Clinical

- BUPA Health Check
- PSA 5.8 (normal range 0-4)
- DRE normal
- MRI – PI-RADS 2 (cancer unlikely)
- Biopsy right lobe cancer GS 3+3
- cT1 N0 M0

Treatment

- Treatment options discussed
 - Surveillance
 - Radiotherapy
 - Robotic nerve sparing surgery
- Robotic prostatectomy
- pT2 Nx margin clear GS 3+3
- Post op PSA <0.1

Treatment Options based on Risk



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Early-stage prostate cancer aged 64yr

- PSA 8.2
- MRI scan PI-RADS 4
- Trans-perineal biopsy – cancer both lobes
- GS 3+4 and 3+3
- cT2c N0 M0
- CPG 2 Low Intermediate risk
- Surveillance, Radiotherapy, Brachytherapy or Surgery!
- How to choose?

Cambridge Prognostic Group recommended by NICE

Cancer Risk based on T stage, PSA and Gleason grade

CPG	T stage		PSA		Gleason Grade	Gleason Group
1	T1	AND	<10	AND	6	1
2	T1-2	AND	10-20	OR	3+4	2
3	T1+2	AND	10-20	AND	3+4 4+3	2 3
4	T3	OR	>20	OR	4+4	4
5	T3 T4	AND/OR OR	>20	AND/OR OR	4+4 9-10	4 5

Treatment Options

Risk group, Patient fitness and Patient Preference

Risk -CPG	Active Surveillance PSA/MRI	Brachytherapy Radiotherapy	Surgery Robotic Nerve sparing	Androgen Deprivation Therapy	AdjuvantCh emo
CPG 1 (low)	+	Patient choice	Patient choice		
CPG 2 (low int)	+	Patient choice	Patient choice		
CPG 3 (high int)	Short term	Brachy or RT	+	Brachy	
CPG 4 (high)	-	RT RT + Brachy	+	+	+/-
CPG 5 (high)	-	RT RT + Brachy	+	+	+

Patient Choice

Adverse effects and cure rates

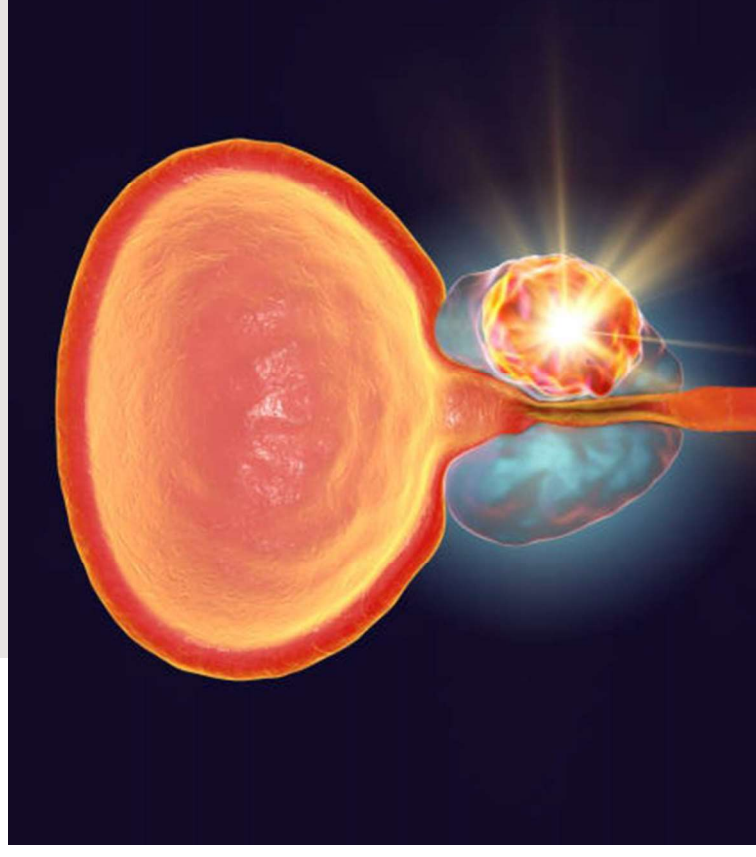
Adverse effects	Active surveillance	Brachytherapy	Radiotherapy	Surgery	Systemic Therapy
Anxiety	+				
Erectile dysfunction		+	++	++	
Urinary		+	+	++	
Rectal		+	++	+	
Short term		++	++	++	++
Cure	May avoid Rx Risk progression to incurable	Higher local failure 15% in 10 yr Can be salvaged	++	++	Adjuvant high risk improves life expectancy

Outcomes by treatment

Treatment	Survival at 10yr	Progression local or distant sites	Rate distant metastases	Long term
Surveillance	98%	21%	8%	Not safe CPG 4/5
Radical radiotherapy	99%	8%	3%	Bowel 10% v 3%
Surgery	99%	8%	3%	Urinary 13% v 8% Sexual 50% v 40%

summary

- 1 Diagnosis
- 2 MRI and reporting systems
- 3 Cambridge Prognostic Groups
- 4 Staging: clinical v pathological
- 5 Treatment options + selection





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Cancer in the workplace

Rehabilitation & Health Management Service Unum



Today's agenda

- 1 What is cancer and its treatment?
- 2 How the condition and treatment may impact on an individual
- 3 Supporting staff at work
- 4 Key points to take away

Statistics on work and cancer

- In 2023, an estimated **3 million people** of working age are living with cancer¹
- This is set to increase to **4 million** by 2030 and **5 million** by 2040.¹
- For those who choose to, work can play a **vital role – 87% of people** employed when diagnosed with cancer say it was important to continue working after their diagnosis²
- A good relationship between an employer and employee is more likely to lead to a **successful return to work**
- Employees and employers **encounter barriers** because the right support is not available

¹ [PowerPoint Presentation \(macmillan.org.uk\)](https://www.macmillan.org.uk)

² [working-through-cancer_tcm9-341781.pdf \(macmillan.org.uk\)](https://www.macmillan.org.uk/tcm9-341781.pdf)

Reasons are
complex

Lack of regular and
meaningful
communication

Lack of shared
understanding from
both sides

“Triple Whammy”
Effect



Addressing the barriers

Employers feel
they lose out

Employees feel
they lose out

Employers want
to do the right
thing

Importance of the Line
Manager

Rigid return to
work policies

Findings: Challenges and solutions



Taking time off for appointments



Offering reduced or flexible hours



Feeling tired at work



Working from home



Feeling distracted at work



Providing counselling or emotional support

Self – insured individuals



Consider
the role
the
insurer
can play

Return to
work
services

Additional
counselling
and advice

What RTW
options
are there?

Is there
potential
for the
flexibility?

Communication is key

Before, during and after treatment.

What might we want to consider before treatment commences?



2

How the condition and treatment may impact on an individual

Quiz

- 1) How many **different types of cancer** do you think exist?
- 2) **True or False:** cancer occurs as a result of abnormal cell division
- 3) **True or False:** 20% of cancers occur due to hereditary/genetic factors
- 4) Name **two potential risk factors** for developing certain types of cancer

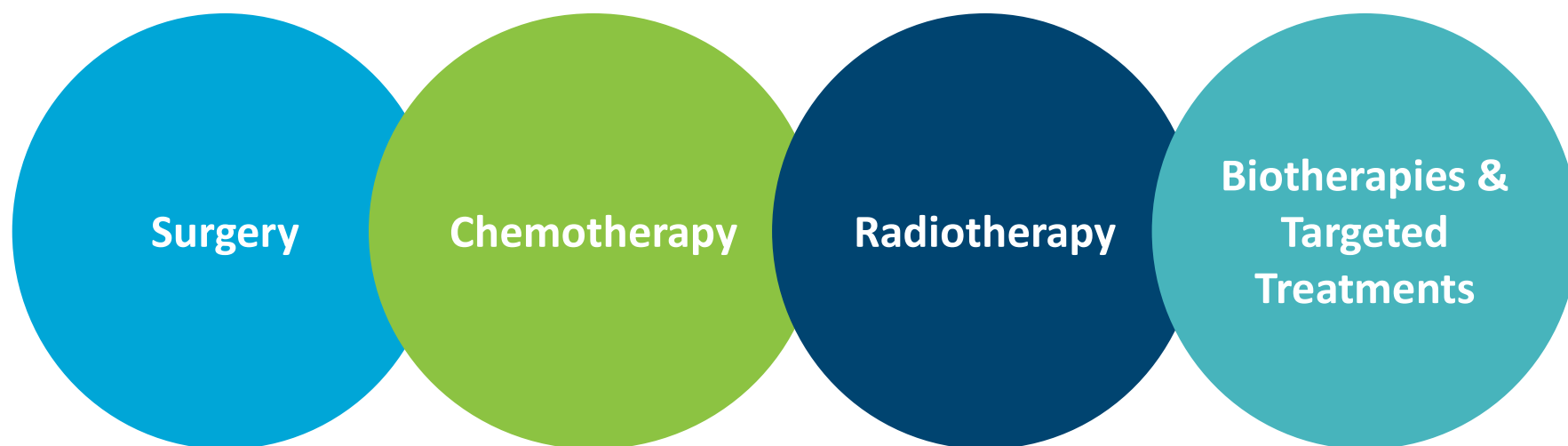


Treatment and side effects



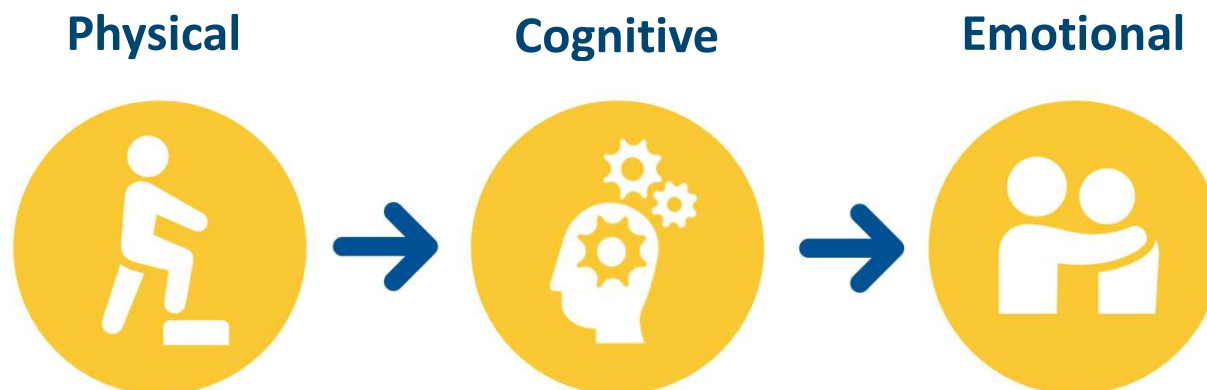
Treatment

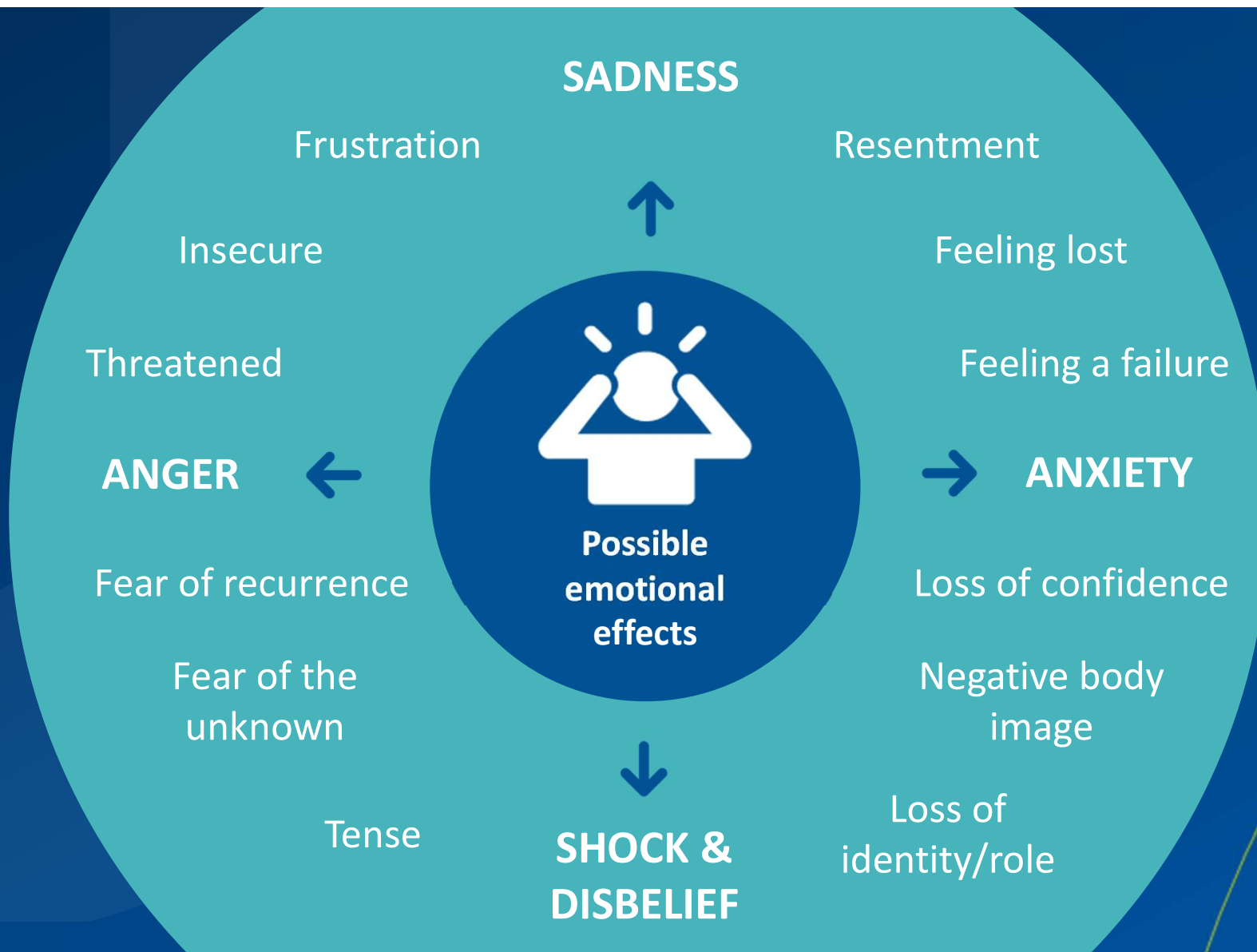
Four main treatments



These may be given in isolation or as combined treatments

Common side effects of treatment





Fatigue

Over **80%**
experience fatigue
while receiving
chemotherapy or
radiation therapy. ¹

It affects people
differently and can be
impact before, during
or after treatment. ²

The cause of cancer-
related fatigue (CRF)
are unknown and have
been previously linked
to many factors. ²



¹ [Fatigue and Cancer Treatment - NCI](#)

² [Tiredness \(fatigue\) and cancer | Macmillan Cancer Support](#)

Communication is key

Before, **during** and after treatment.

What might you consider during an individual's treatment?





3

Supporting staff at work

Fictional case study: James

Job Information: Project Manager, full time, office based, been in the role for 5 years

Absent for: 8 months

Graduated return to work duration: 4 months

Nature of illness: Metastatic prostate cancer

Treatment: Palliative hormone treatment and radiotherapy

Symptoms:

- Fatigue
- Depleted concentration
- Urinary frequency and urgency
- Poor concentration

Current function:

- Motivated to return to work
- Gets tired very easily when doing household chores
- Advised by doctor not to drive due to strength of pain relief



1. How would you support James before treatment?
2. How would you support James during treatment?
3. How would you support James after treatment?

Supporting a return to work

Early intervention

To support employees wanting to work through their treatment



Plan a graduated return to work

Staged introduction

Appropriate initial tasks

Suitable adjustments

Monitoring and reviewing progress

Support

Dedicated contact

Collaborative working

Discuss and provide further support



Communication is key

Before, during and **after** treatment.

What might we want to think about when treatment is complete?



Agenda

4

Key points to take away

Key points to take away



Many
different
types of
cancer and
treatment

Side effects
vary and can
be both short-
term and
long-term

Physical,
emotional and
cognitive side
effects are
significant

Fatigue is the
most
common side
effect of
cancer
treatment

Communicati
on is key



Thank you

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Paddling in invested waters ?

Cancer screening – should we do it?

Dr John O'Brien

July 2024



Early
identification

Prevention

The Sutton Principle



Why do you rob
banks?

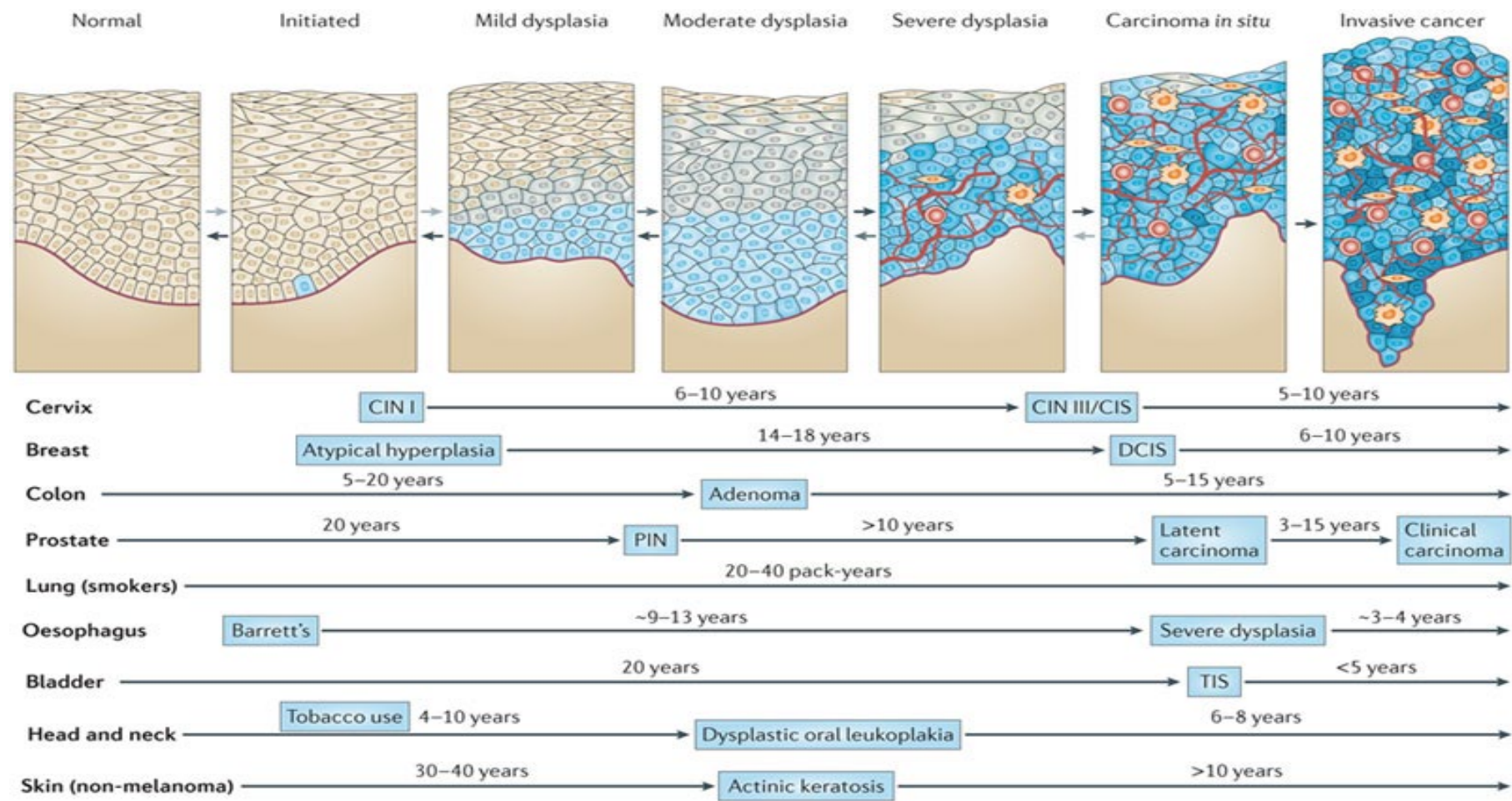
Because that is the
where the money is!

Benefits of cancer
screening

Early Identification

- Small cancers detected
- More likely to be cured with simple treatment

From first development to clinical presentation



Benefits of cancer
screening

Prevention

- Cervical cancer
- Prevention before screening is even better!

Benefits of cancer
screening

Prevention

Colon cancer with colonoscopy

Not all cancer is progressive and life-threatening

Concepts

- Benign vs Malignant
- Indolent vs Aggressive
- Indolent lesion of epithelial origin IDLE*
- Indolent disease might account for 15–75% of all cancers*
- Localised growth vs early spread
- Biology vs stage

**Lancet Oncol.* 2014 May ; 15(6): e234–e242. doi:10.1016/S1470-2045(13)70598-9.

Published in final edited form as:

Lancet Oncol. 2014 May ; 15(6): e234–e242. doi:10.1016/S1470-2045(13)70598-9.

Addressing overdiagnosis and overtreatment in cancer: a prescription for change

Prof. Laura J. Esserman, MD, Prof. Ian M. Thompson, MD, Prof. Brian Reid, MD, Prof. Peter Nelson, MD, Prof. David F. Ransohoff, MD, Prof. H. Gilbert Welch, MD, Shelley Hwang, MD, Prof. Donald A. Berry, PhD, Prof. Kenneth W. Kinzler, PhD, Prof. William C. Black, MD, Prof. Mina Bissell, PhD, Howard Parnes, PhD, and Sudhir Srivastava, PhD

Not all cancer is progressive and life-threatening

Perspectives > Second Opinions

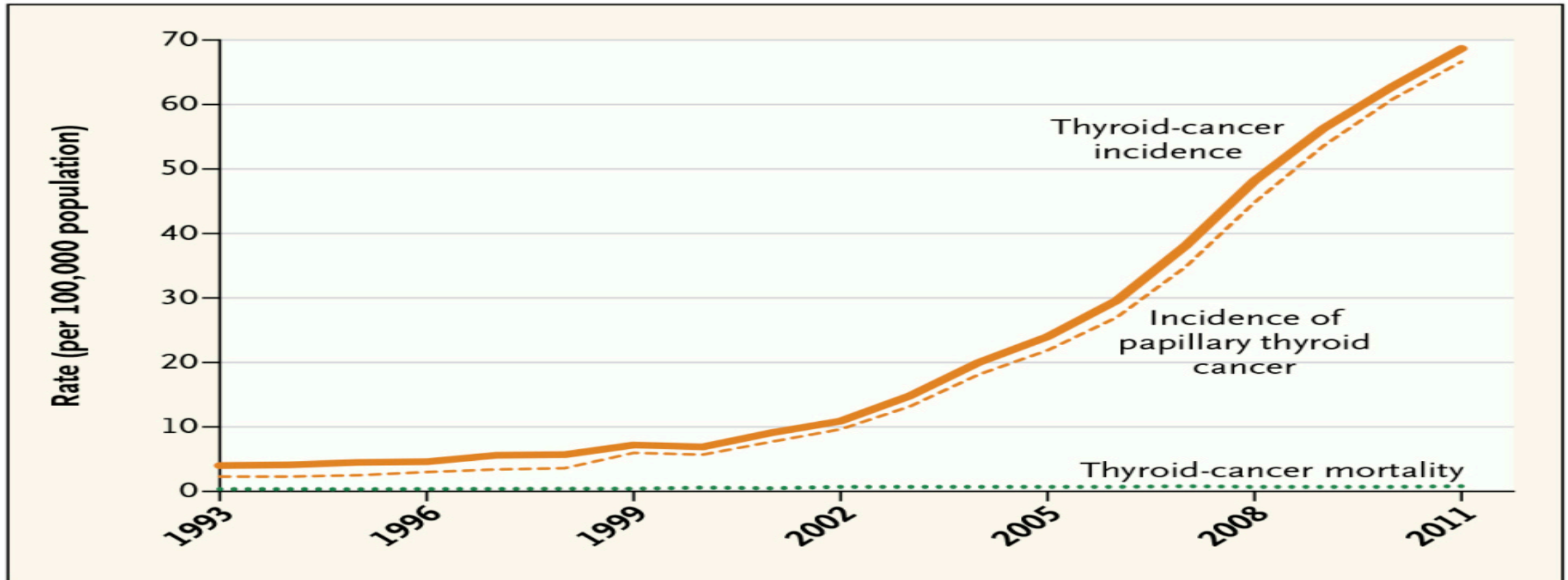
'We Caught It Early': Blessing or Burden?

— Cancer screening has its drawbacks

by Shannon Casey, PA-C January 31, 2024

... Howard V. Esserman, MD, Prof. Ian M. Thompson, MD, Prof. Brian Reid, MD, Prof. Peter Nelson, MD, Prof. David F. Ransohoff, MD, Prof. H. Gilbert Welch, MD, Shelley Hwang, MD, Prof. Donald A. Berry, PhD, Prof. Kenneth W. Kinzler, PhD, Prof. William C. Black, MD, Prof. Mina Bissell, PhD, Howard Parnes, PhD, and Sudhir Srivastava, PhD

Thyroid cancer incidence



Thyroid-Cancer Incidence and Related Mortality in South Korea, 1993–2011.

Data on incidence are from the Cancer Incidence Database, Korean Central Cancer Registry; data on mortality are from the Cause of Death Database, Statistics Korea. All data are age-adjusted to the South Korean standard population.

Thyroid cancer

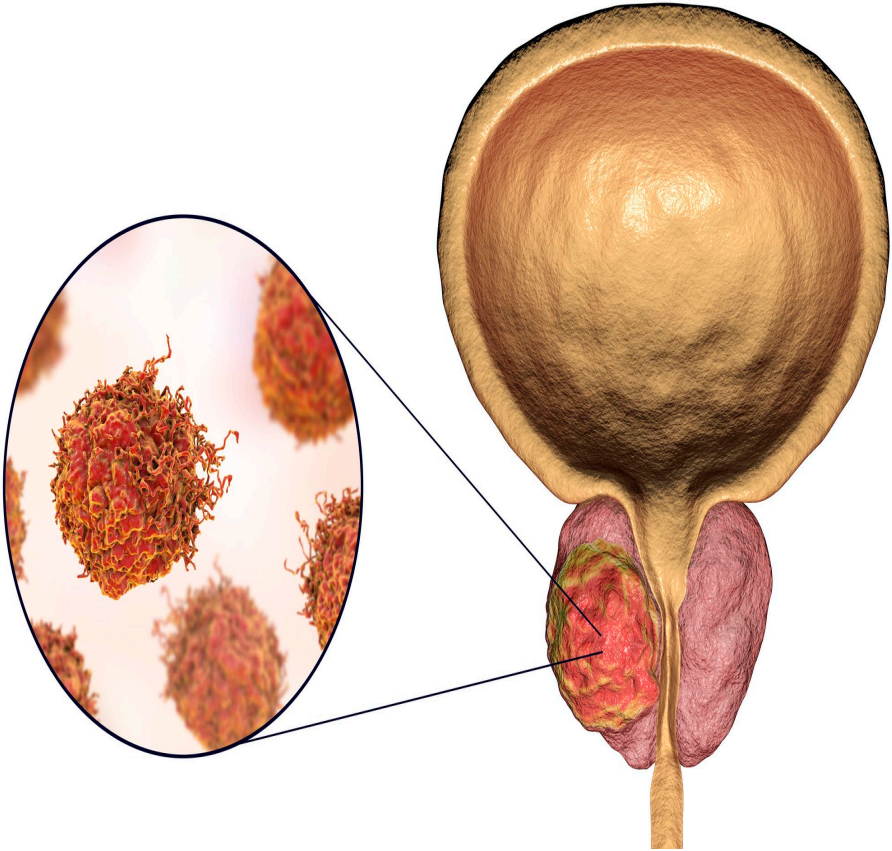
- 36% had at least 1 papillary carcinoma at autopsy
- Calculation that “if sectioned finely enough, virtually every person would be found to harbor a thyroid cancer.”
- SEER data indicate about half the increase is due to tumors 1 cm or smaller and 87% smaller than 2 cm

Prostate cancer

➤ Unscreened men autopsy study

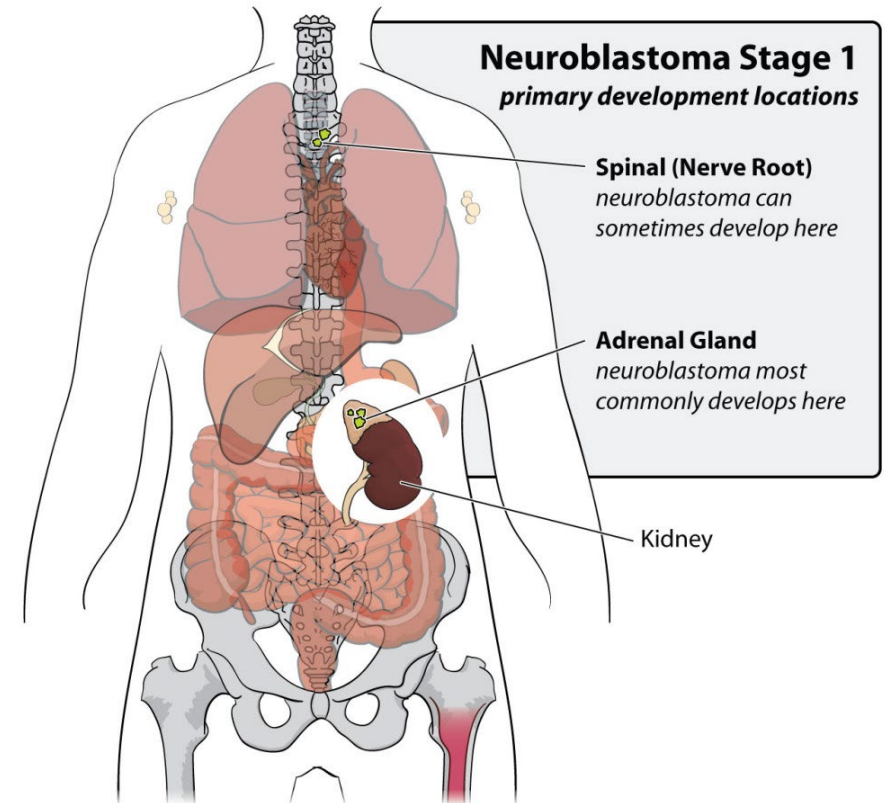
Age	%
20 – 30	3.6
31 – 49	8.9
41 – 50	14.0
51 – 60	24.0
61 – 70	31.0

➤ Over 70 incidence rises to approximately 70%



Neuroblastoma

- Childhood cancer
- Screening by urinary catecholamines
- Study in Quebec showed not effective and harmful
- Neuroblastoma S – spontaneously regress



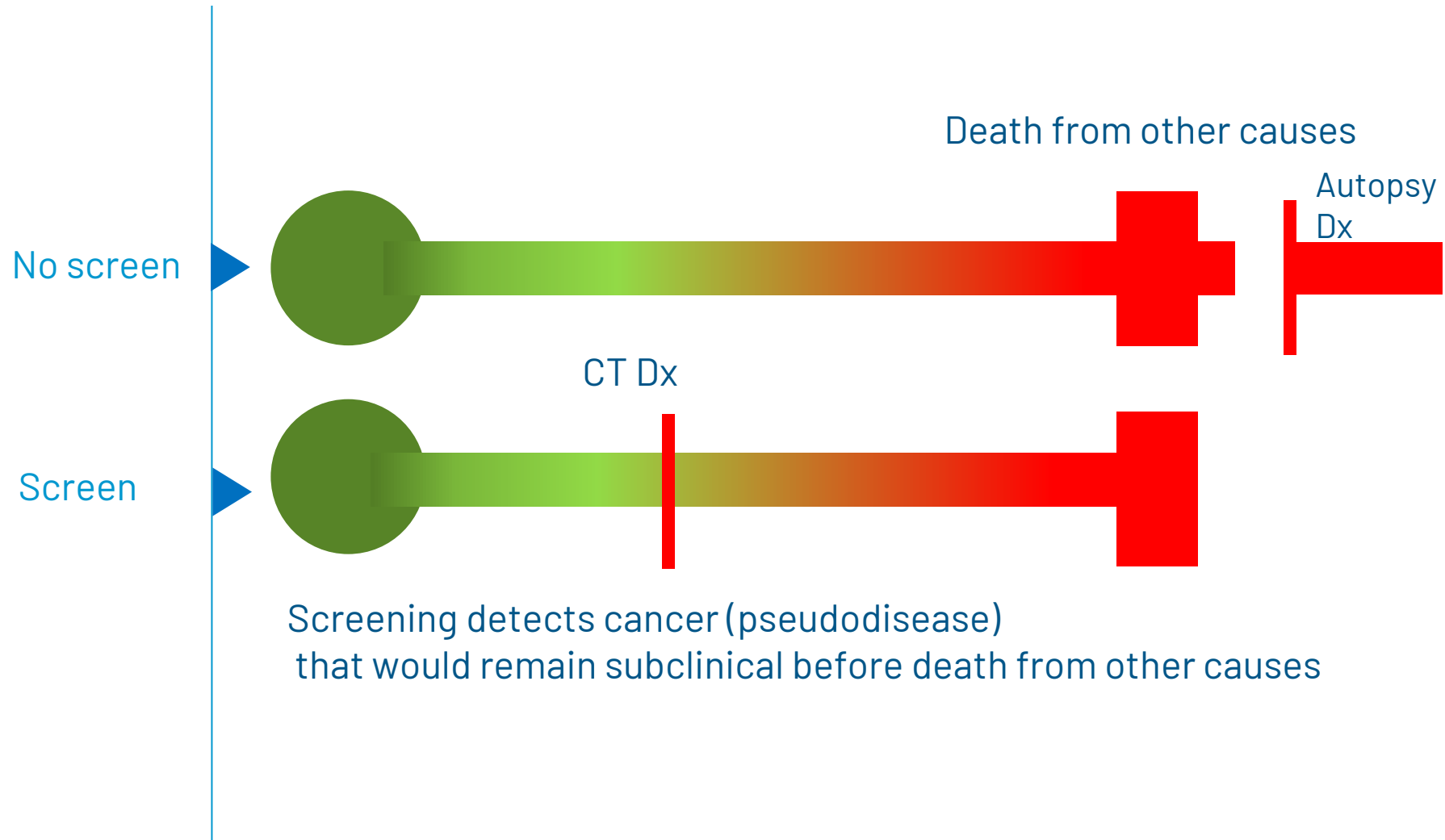
Overdiagnosis

- Would not go on to cause symptoms or death
- Never progresses
- Regresses
- Subject dies from other causes before cancer is symptomatic

False-positive

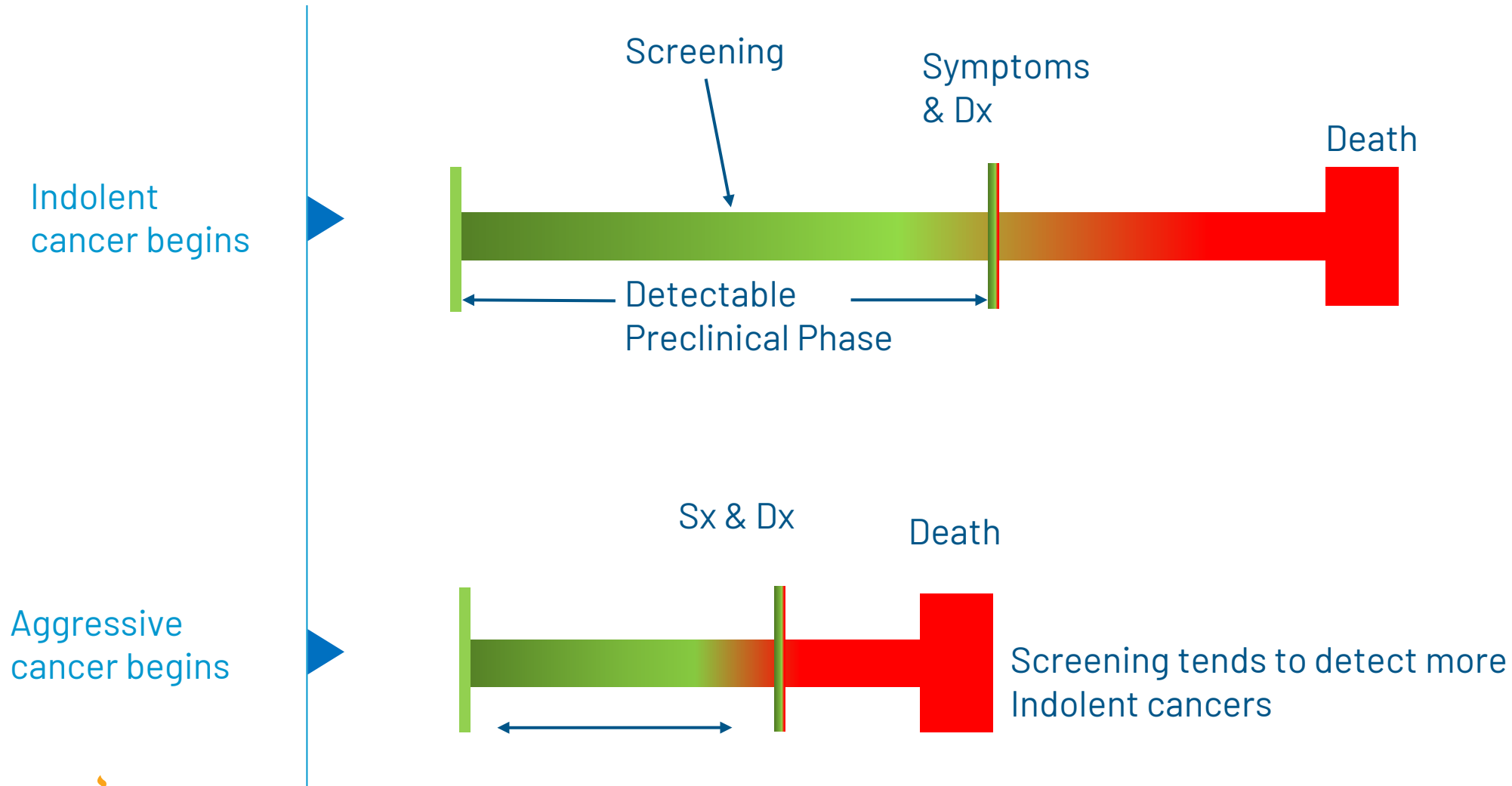
Cancer Screening

Overdiagnosis Bias (Pseudodisease)



Cancer Screening

Length bias



Cancer screening

No Screening

500 people with
Progressive cancer

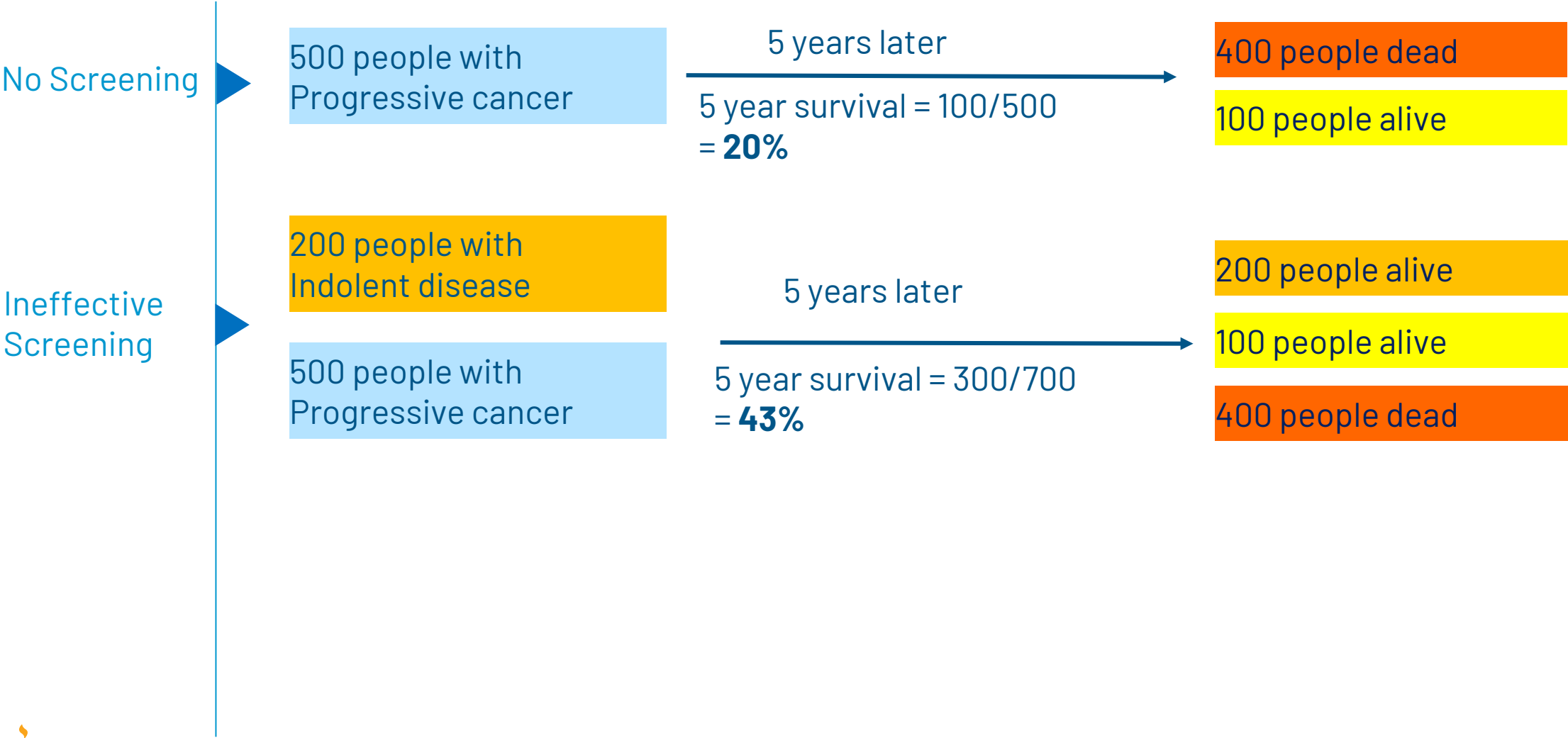
5 years later

5 year survival = $100/500$
= **20%**

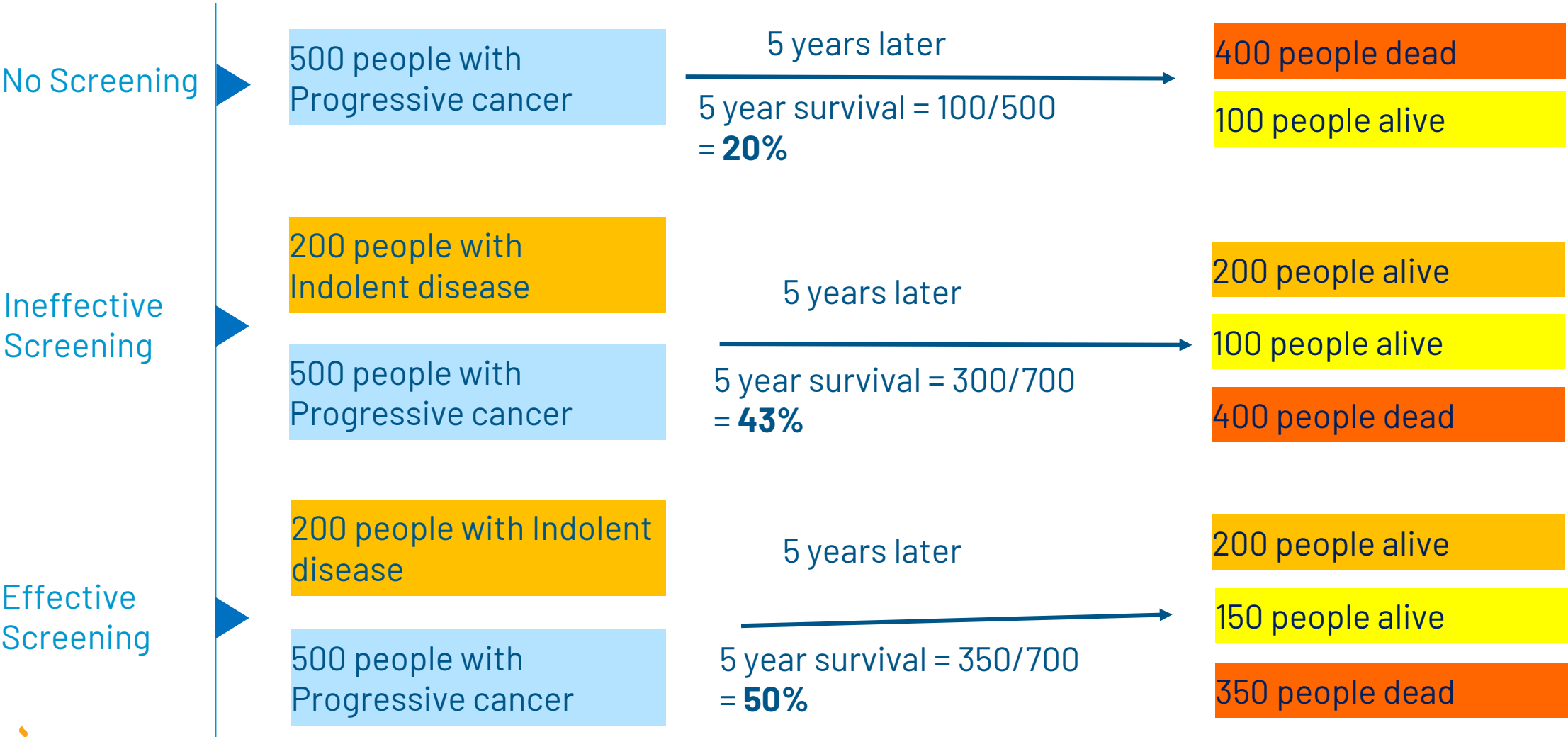
400 people dead

100 people alive

Cancer screening



Cancer screening

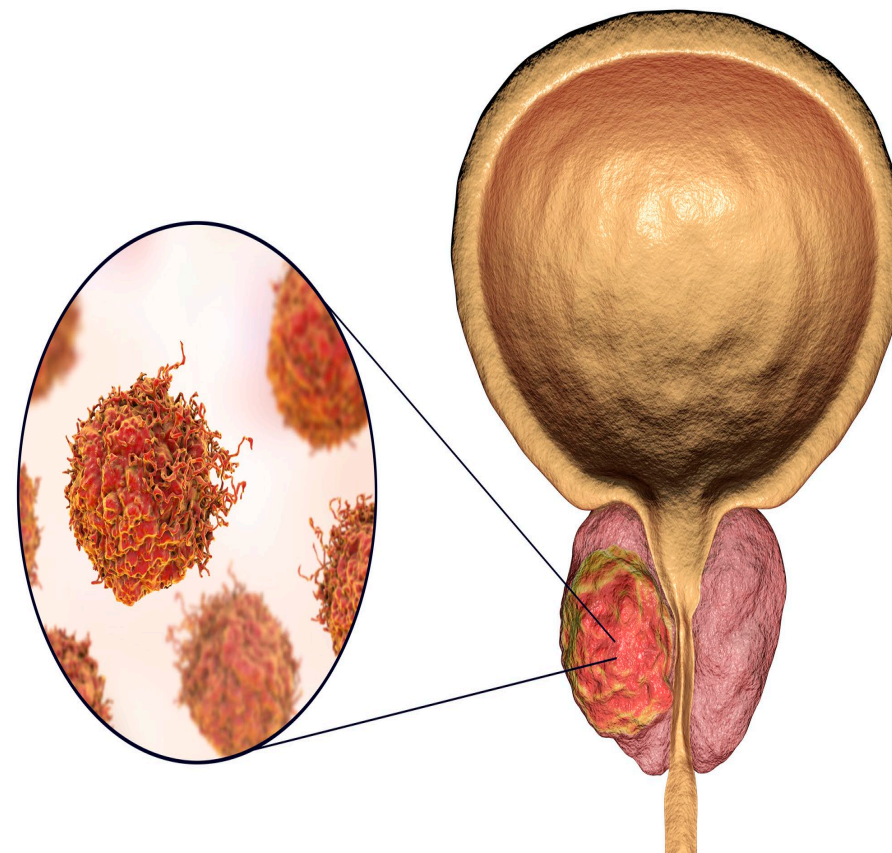




Prostrate Cancer

Prostate cancer

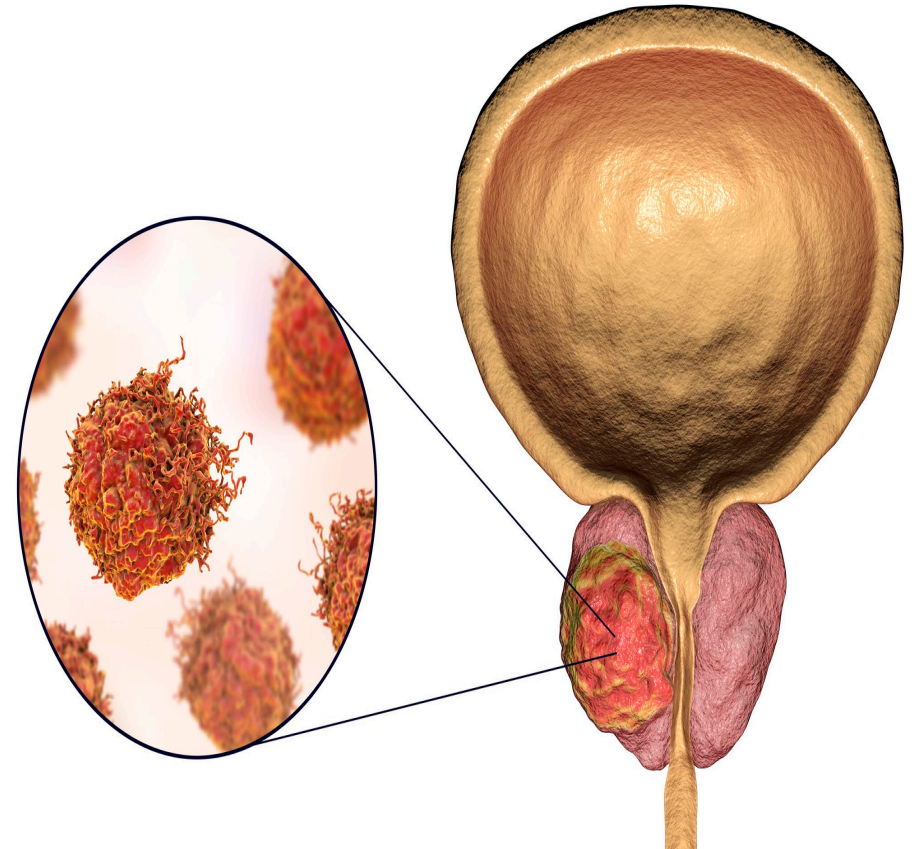
PSA screening may reduce prostate cancer mortality risk but is associated with false-positive results, biopsy complications, and overdiagnosis. Compared with conservative approaches, active treatments for screen-detected prostate cancer have unclear effects on long-term survival but are associated with sexual and urinary difficulties



JAMA. 2018;319(18):1914-1931. doi:[10.1001/jama.2018.3712](https://doi.org/10.1001/jama.2018.3712)

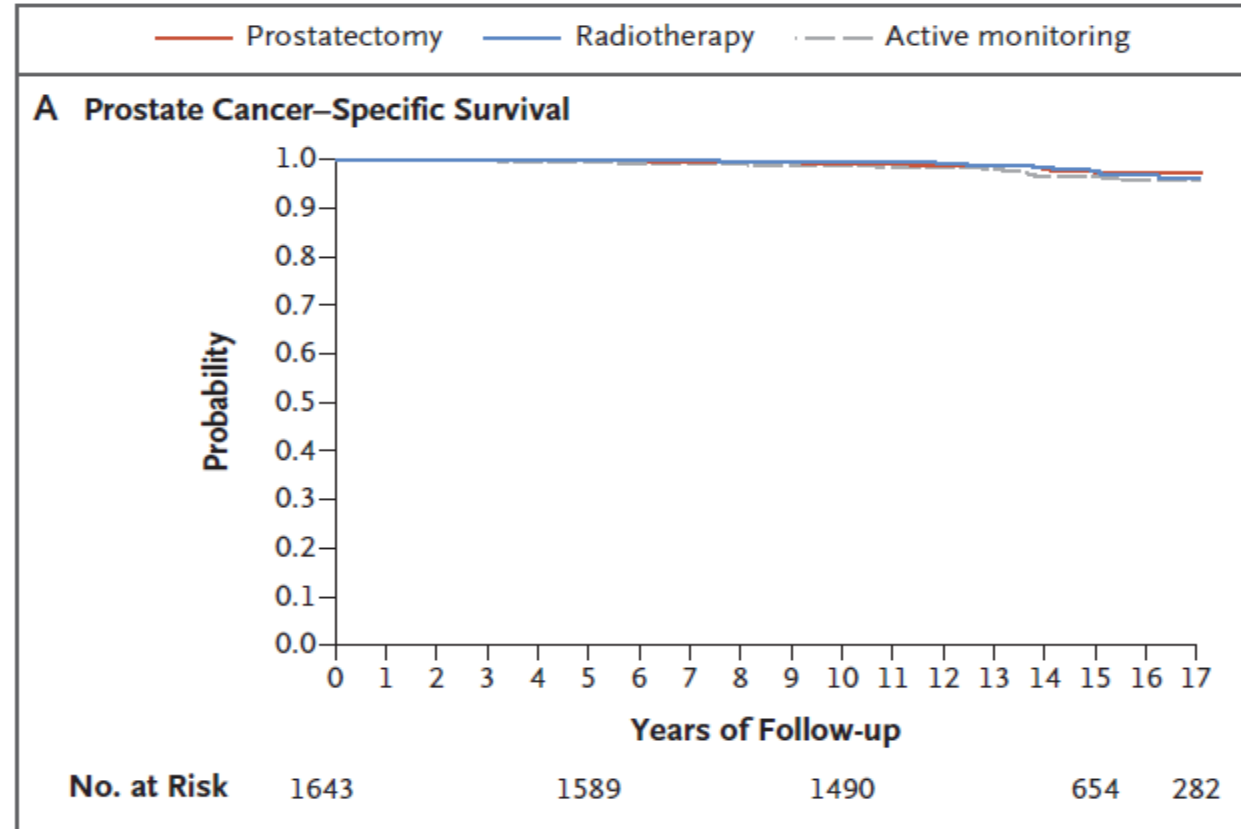
Prostate cancer – Advances in detection

- Total and free
- PHI prostate health index Total, free and p2PSA
- MRI
- Urine testing for genetic markers potential to distinguish high risk from indolent prostate cancer
- Urine tests for microscopic haematuria marketed to public



JAMA. 2018;319(18):1914-1931. doi:10.1001/jama.2018.3712

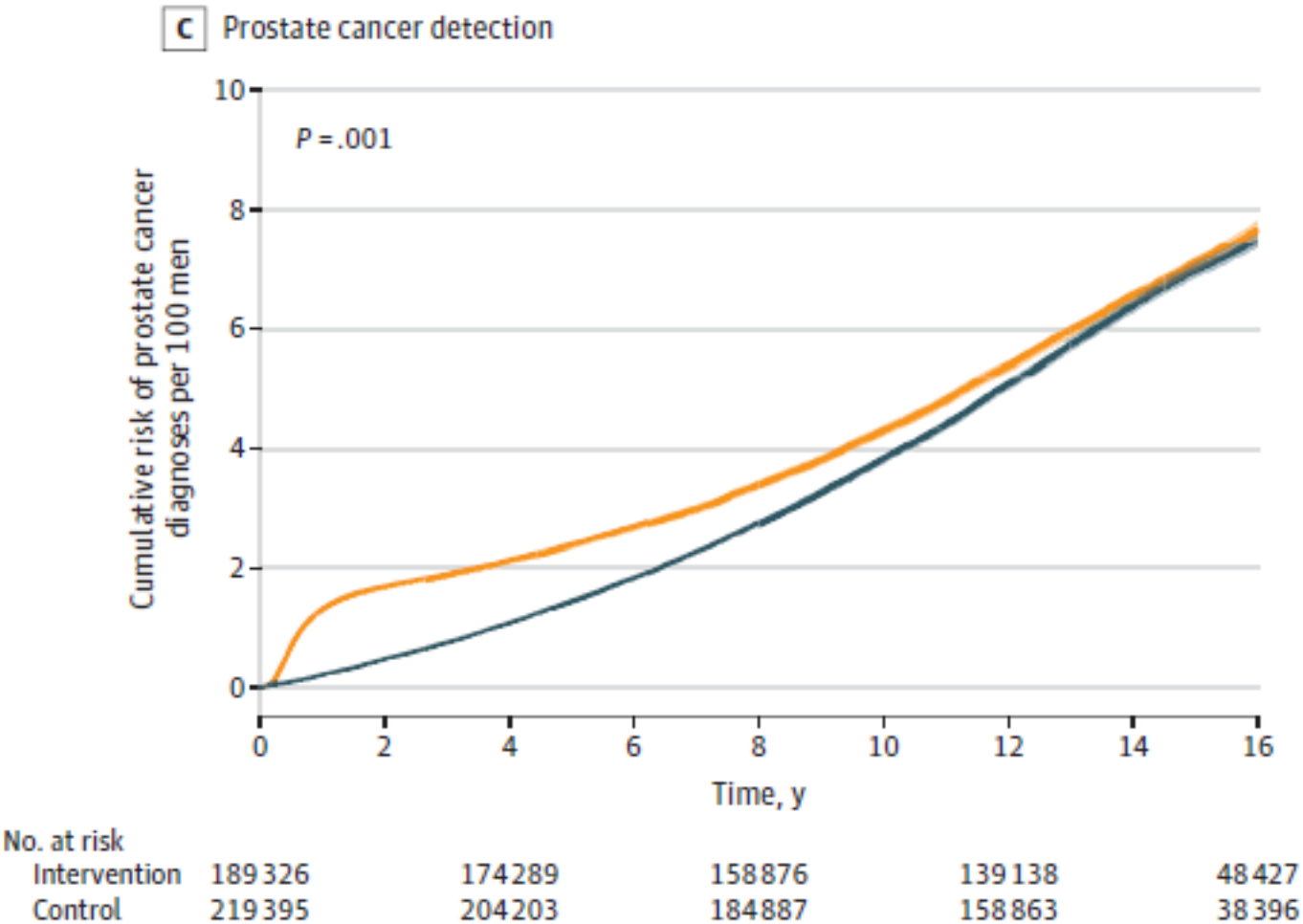
Prostate cancer



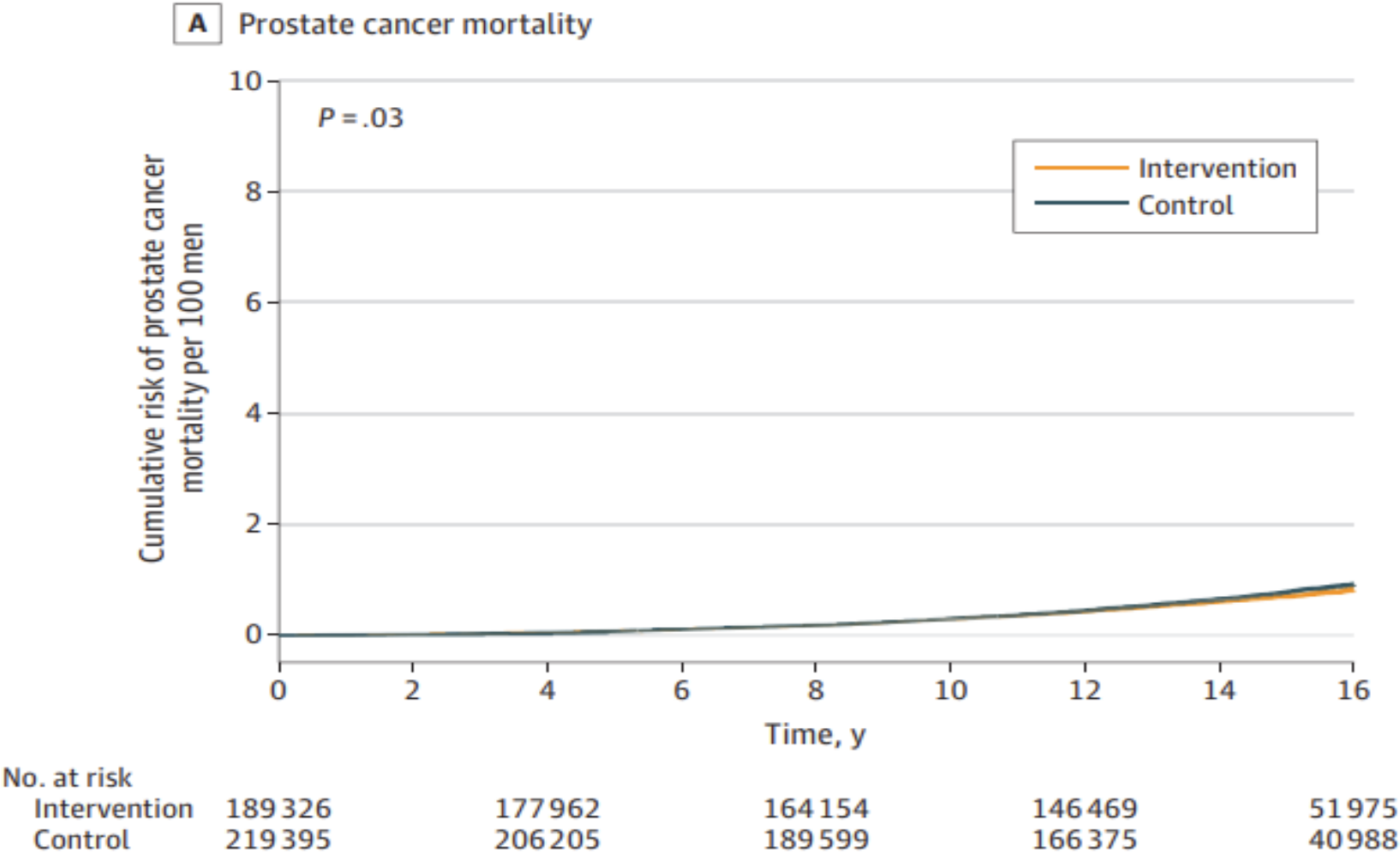
Hamdy FC et al. *N Engl J Med* 2023;388:1547-1558



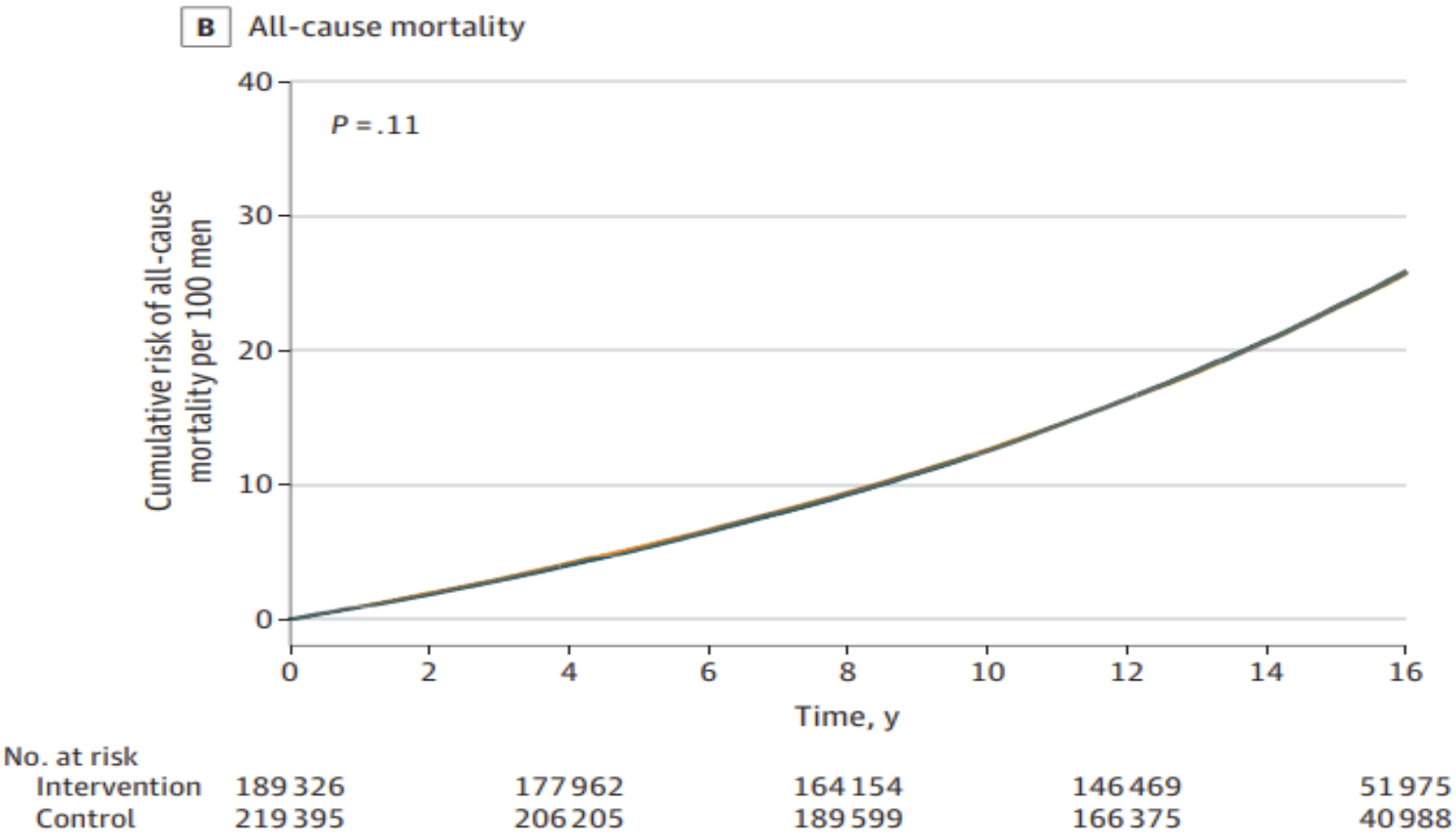
Effect of the Trial Intervention on the Cumulative Incidence (95% CI) of Prostate Cancer Mortality and Diagnosis and All-Cause Mortality After a Median 15-Year Follow-Up



Effect of the Trial Intervention on the Cumulative Incidence (95% CI) of Prostate Cancer Mortality and Diagnosis and All-Cause Mortality After a Median 15-Year Follow-Up



Effect of the Trial Intervention on the Cumulative Incidence (95% CI) of Prostate Cancer Mortality and Diagnosis and All-Cause Mortality After a Median 15-Year Follow-Up



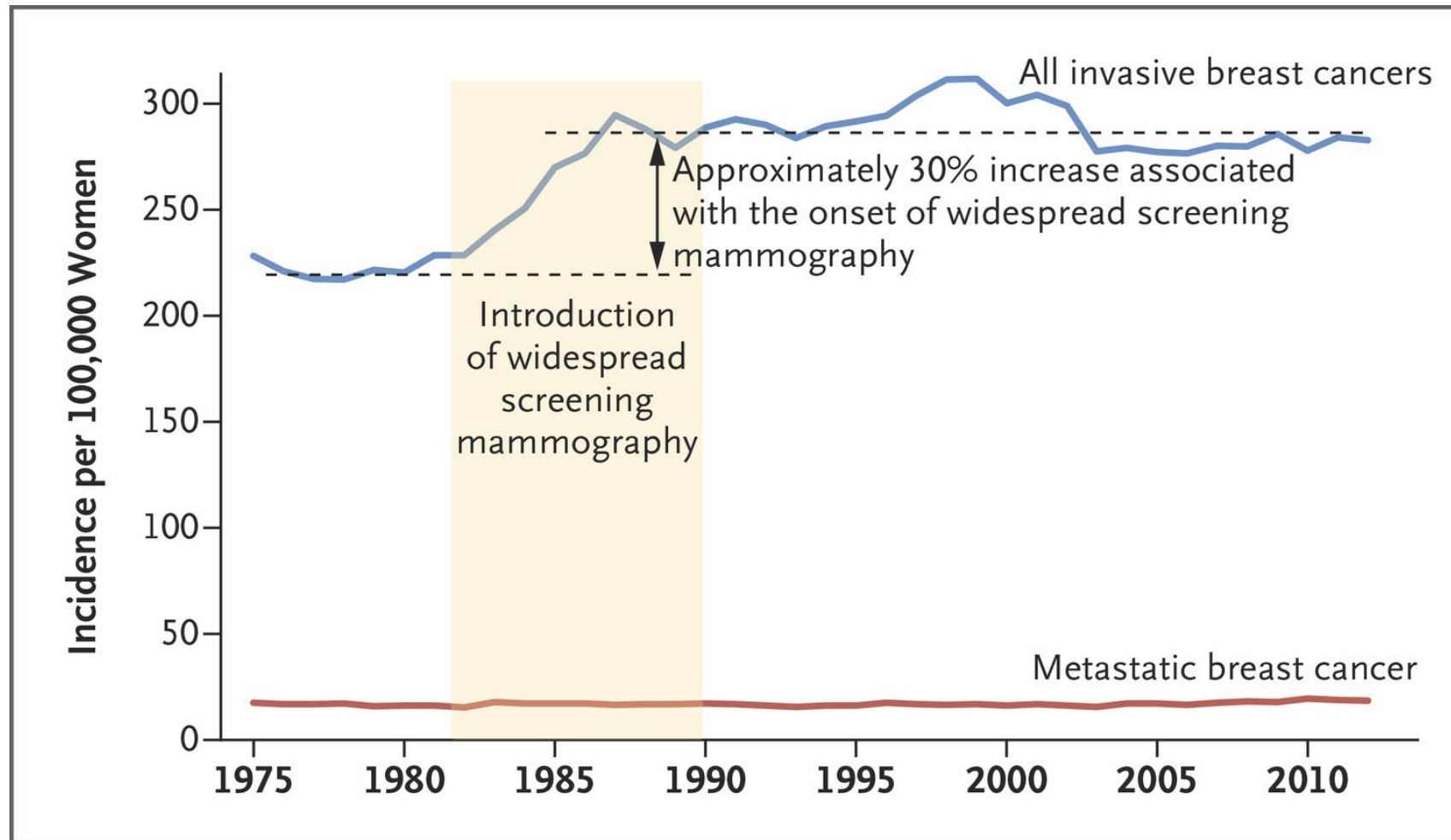


Breast Cancer

Breast cancer screening – Self examination

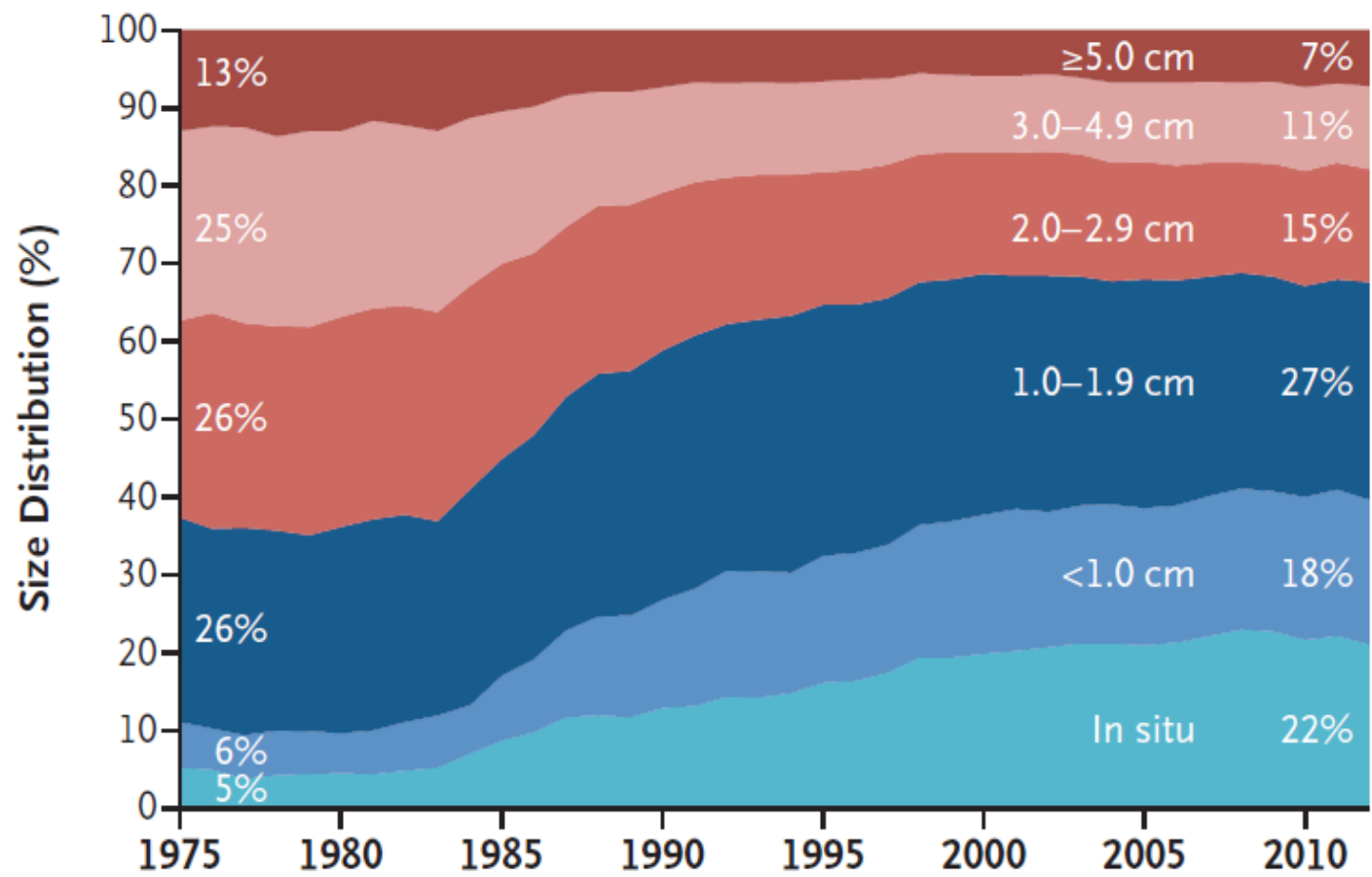
- Reduces size of tumour at diagnosis
- Does not impact on mortality
- Not recommended as a screening procedure

Temporal Relationship between the Introduction of Screening Mammography and Increased Incidence of Invasive Breast Cancer.



Breast-Cancer Tumor-Size Distribution and Size-Specific Incidence among Women 40 Years of Age or Older in the United States, 1975–2012.

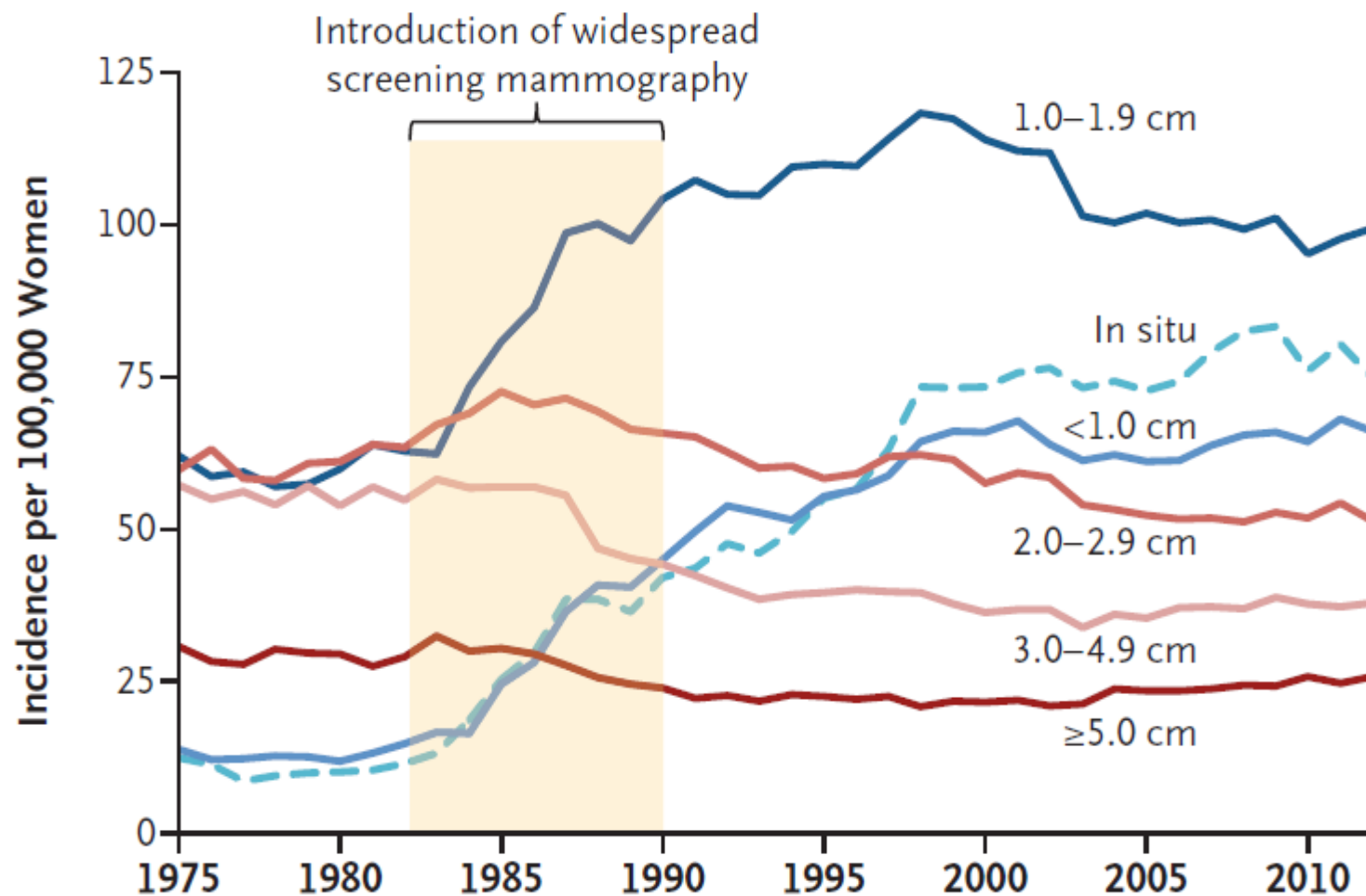
A Tumor Size Distribution



Welch HG et al. N Engl J Med 2016;375:1438–1447.

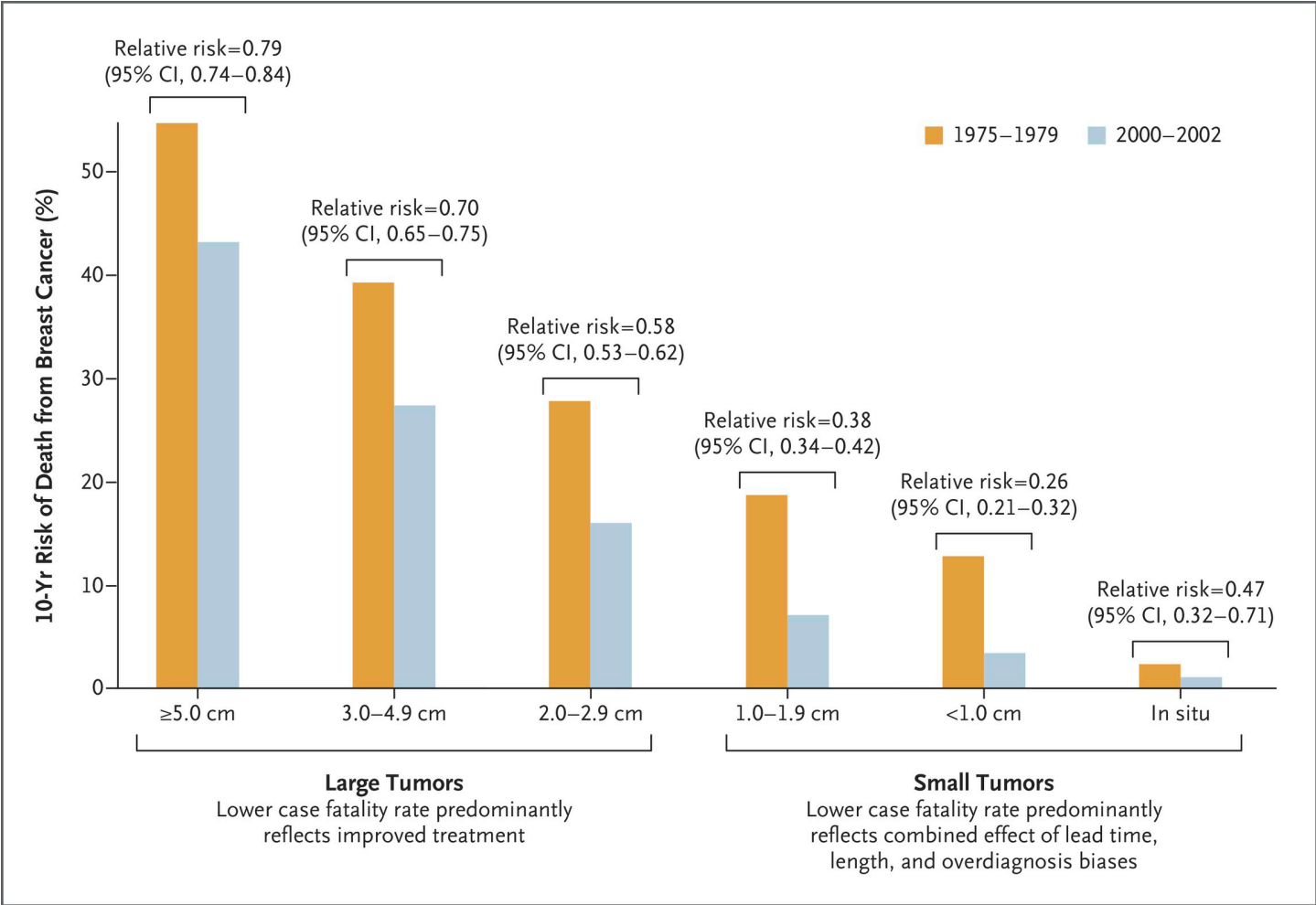
Breast-Cancer Tumor-Size Distribution and Size-Specific Incidence among Women 40 Years of Age or Older in the United States, 1975–2012.

B Size-Specific Incidence



Welch HG et al. *N Engl J Med* 2016;375:1438–1447.

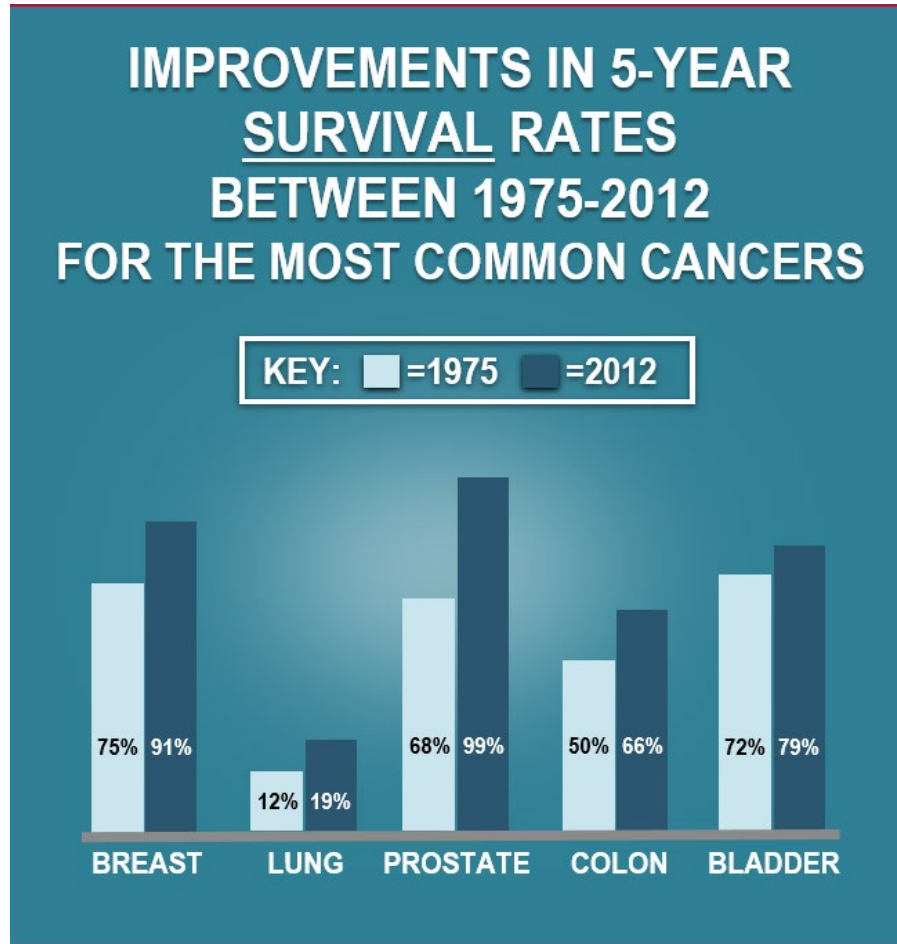
Change in Size-Specific Case Fatality Rate





Lung Cancer

Lung cancer screening – Precision selection vs precision technology



seer.cancer.gov

Lung cancer screening – Precision selection vs precision technology

- Original studies with chest radiograph (chest x-ray) and sputum cytology 1980's
- No survival benefit
- PLCO (Prostate, Lung, Colorectal and Ovarian trial)
- **Confirmed that annual chest radiograph has no impact on lung cancer diagnosis or mortality**

National Lung Screening Trial – NSLT

55 to 74 yrs, > 30P-Y, active or stop<15 yrs

	n	« Positive » (diam≥4mm)	Lung cancers	Lung cancer deaths	Risk reduction
Low-dose CT	26.455	18.146 (68.5%)	1.060	346 (1.3%)	20,0%
Chest X-ray	26.232	5.043 (19.2%)	941	425 (1.6%)	

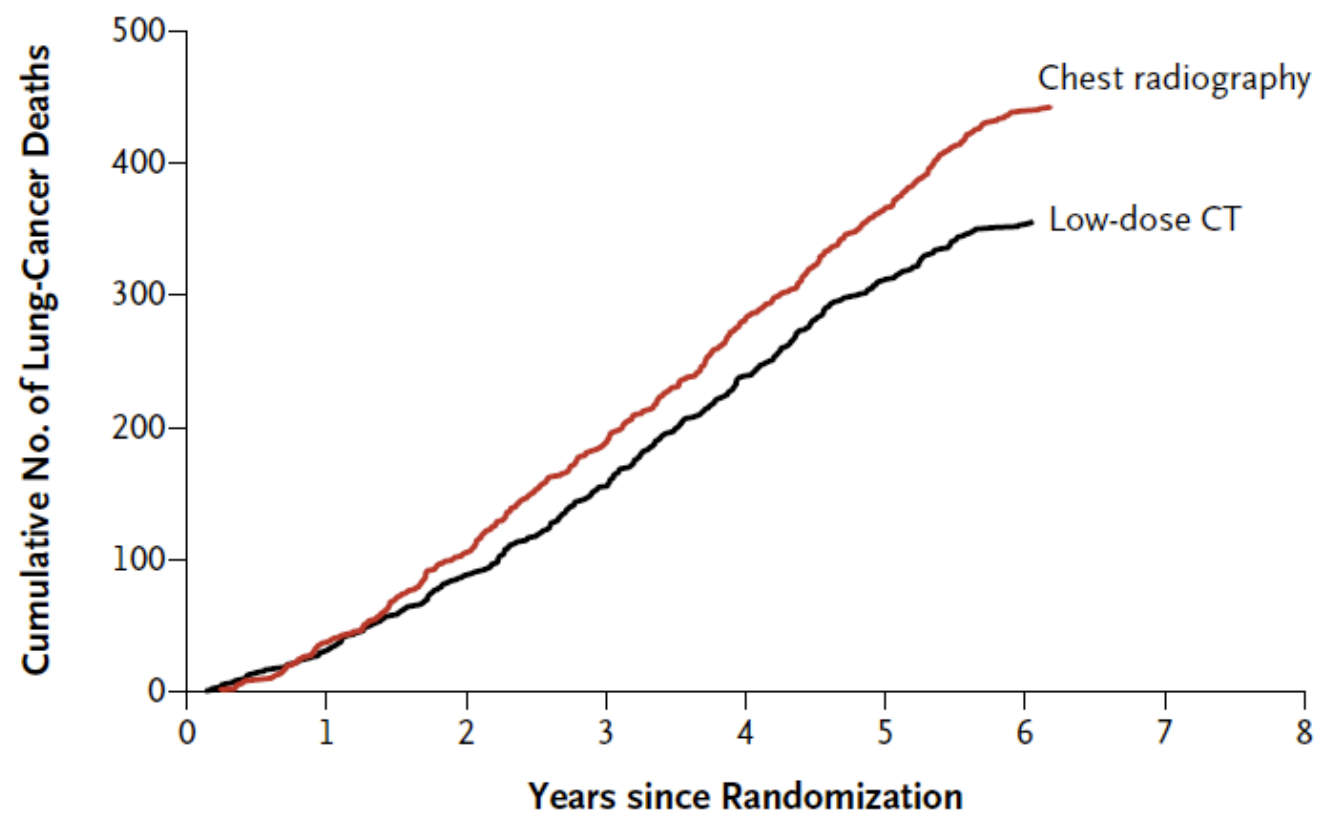
NLST, NEJM 2011; 365(5):395-409

NNS to prevent 1 lung cancer death = 320...

National Lung Screening Trial – NSLT

Deaths from lung cancer

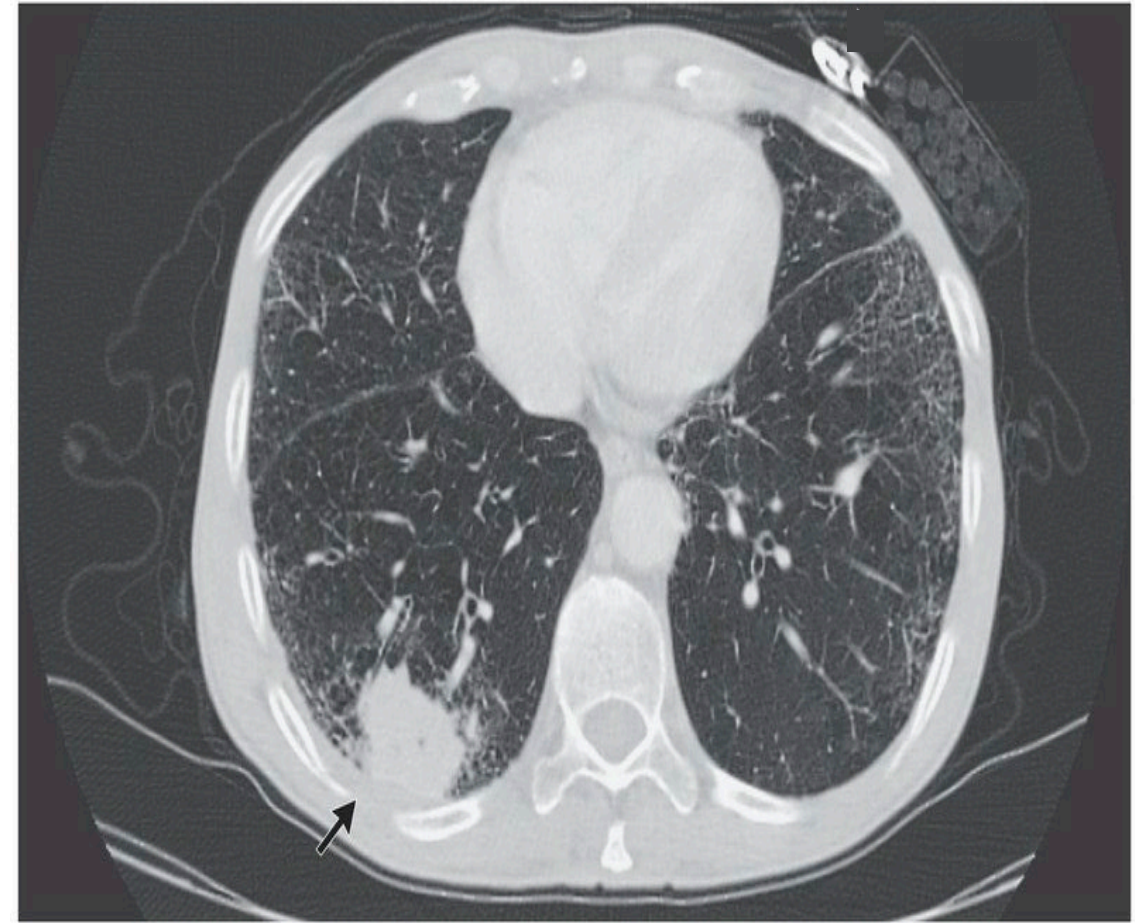
B Death from Lung Cancer



NLST, NEJM 2011; 365(5):395-409

Low dose CT scanning for Lung cancer

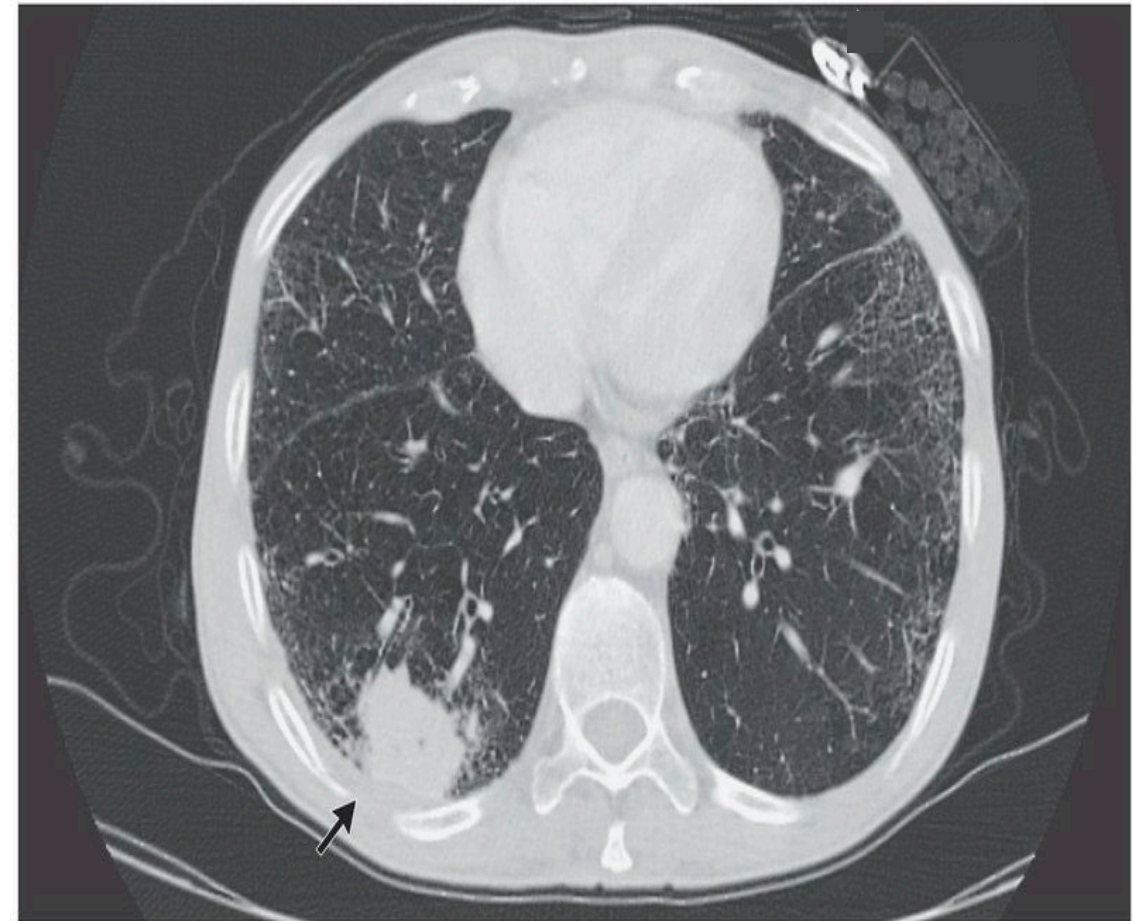
- At risk population criteria have expanded
- NHS- People aged 55 to 74 with a GP record including a history of smoking will be assessed and invited for screenings and smoking cessation services
- 70% of the screening took place in mobile units parked in convenient places - such as supermarket car parks - to ensure easy access and focused on more deprived areas where people are 4 times more likely to smoke.



L Conti, S Gatt. N Engl J Med 2018;379:e17.

Low dose CT scanning for Lung cancer

- During the initial phase almost 900,000 people were invited for checks, 375,000 risk assessments made and 200,000 scans were carried out.
- More than 2,000 people were detected as having cancer, 76% at an earlier stage compared to 29% in 2019 outside of the programme.



L Conti, S Gatt. N Engl J Med 2018;379:e17.

Multi cancer early detection – Liquid biopsy

- In 1869, Thomas Ashworth described the microscopic observation of circulating tumor cells (CTC) in the blood of a man with metastatic cancer. He concluded that the CTC must have passed through the circulatory system to arrive at the vein from which the blood was collected
- Many techniques developed for detecting circulating cancer cells
- More recently focus has been on circulating Tumour DNA and proteomics
- Galleri and CancerSEEK

Multi-cancer blood test shows real promise in NHS study

Multi-cancer blood test shows real promise in NHS study - BBC News



Multi-cancer blood test in NHS

An update on the ongoing NHS-Galleri trial

📅 29 May 2024



Professor Peter Johnson

Cancer

Based on data from the first year of the three-year [NHS-Galleri trial](#), NHS England has decided that it will wait to see final results, expected in 2026, before considering whether an NHS rollout of the Galleri multi cancer early detection test (the Multi Cancer Blood Test Programme, MCBT) should go ahead.

Limitations of Multi-Cancer Detection Tests

Tumor Markers for Screening, Not Diagnosis

Uncertain Clinical Utility for the Unselected Population

1. Sensitivity (false -) and Specificity (false +) issues
2. Limited predictive value in early-stage cancer
3. Must use in combination with standard cancer screening tests
4. Study design limitations > Performance unclear when screening unaffected populations
5. Diagnostic process unclear
6. Dynamic nature of cancer: Repeat tests needed
7. Harms:
 - Anxiety
 - Overdiagnosis, overtreatment
 - Cost vs. impact

Cancer screening trials

End points

- Mortality – all cause vs cancer related
- Stage 3 or 4 cancer

Insurance implications Life products

- Potential for avoiding or delaying cancer death (Lead time bias)
- Underwriting complexity increases

- Earlier diagnosis – financial implications paying years earlier
- Paying for 'incidental or non-serious' disease – pricing implications

Insurance implications Critical Illness



What if we don't screen for cancer

What if we don't screen for cancer?

Research

JAMA Internal Medicine | [Original Investigation](#)

Estimated Lifetime Gained With Cancer Screening Tests A Meta-Analysis of Randomized Clinical Trials

Michael Bretthauer, MD, PhD; Paulina Wieszczy, MSc, PhD; Magnus Løberg, MD, PhD;
Michal F. Kaminski, MD, PhD; Tarjei Fiskergård Werner, MSc; Lise M. Helsing, MD, PhD; Yuichi Mori, MD, PhD;
Øyvind Holme, MD, PhD; Hans-Olov Adami, MD, PhD; Mette Kalager, MD, PhD

JAMA Intern Med. doi:10.1001/jamainternmed.2023.3798
Published online August 28, 2023.



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What if we don't screen for cancer?

Research

JAMA Internal Medicine | [Original Investigation](#)

Estimated Lifetime Gained With Cancer Screening Tests A Meta-Analysis of Randomized Clinical Trials

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Øyvind Holme, MD, PhD; Hans-Olov Adami, MD, PhD; Mette Kalager, MD, PhD

JAMA Intern Med. doi:10.1001/jamainternmed.2023.3798
Published online August 28, 2023.

CONCLUSIONS AND RELEVANCE

The findings of this meta-analysis suggest that current evidence does not substantiate the claim that common cancer screening tests save lives by extending lifetime, except possibly for colorectal cancer screening with sigmoidoscopy.



What if we don't screen for cancer?

Median follow-up

10 years for computed tomography, prostate-specific antigen testing, colonoscopy;

13 years for mammography;

15 years for sigmoidoscopy and FOBT.

The only screening test with a significant lifetime gain was sigmoidoscopy (110 days; 95%CI, 0-274 days).

There was no significant difference following

Mammography (0 days; 95%CI, -190 to 237 days),

Prostate cancer screening (37 days; 95%CI, -37 to 73 days),

Colonoscopy (37 days; 95%CI, -146 to 146 days),

FOBT screening every year or every other year (0 days; 95%CI, -70.7 to 70.7 days),

Lung cancer screening (107 days; 95%CI, -286 days to 430 days).

- Biology of cancers differ considerably
- Not all cancers are aggressive and result in fatal outcomes if untreated
- Improvements in life expectancy resulting from screening are modest or controversial for many cancers
- The challenge is to identify aggressive cancers early
- Prevention is ideal where possible
- Treatment advances are responsible for a large part of the improvement in cancer outcomes

What I said



There is no value in
screening for
cancer

What I didn't say





Thank you

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Thank you

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